

3. The athlete then slaloms through the four cones in the center aisle in both directions before then running to the cone at the opposite end of the third aisle.
4. The athlete sprints the 10 meters of the third aisle to complete the test.
5. The tester will stop the timer once the athlete has completed the third aisle of cones and record the time.
6. *Note:* Normative data for the Illinois test are presented [tables 14.31](#) and [14.32](#) near the end of the chapter.

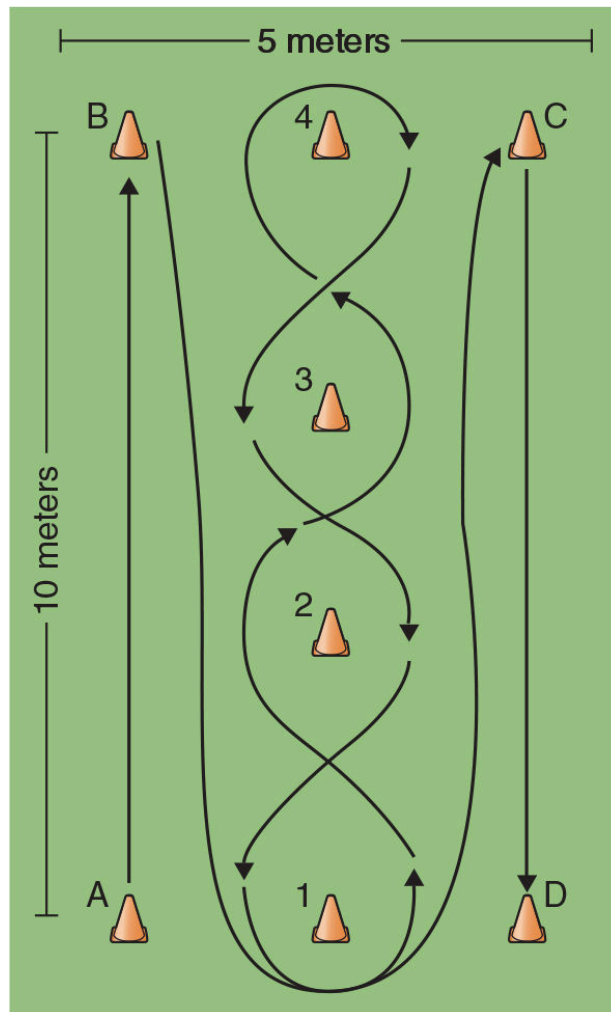


FIGURE 14.25 Layout for the Illinois test.

Reprinted by permission from R. Lockie, "Testing, Assessment, and Monitoring of Agility and Quickness," in *Developing Agility and Quickness*, 2nd ed., edited for the National Strength and Conditioning Association by J. Dawes (Human Kinetics, 2019), 93.

Speed

30 STRAIGHT-LINE SPRINT TESTS

Equipment

- Stopwatch or timing gates, if available, with corresponding software on a laptop, tablet, or smartphone application

- Flat running surface with start and finish lines a specified distance apart (e.g., 40 yd [37 m]; 10 m, 20 m, 40 m), with at least 20 yards (18 m) after the finish line for deceleration
- Where multiple timing gates are available, set these up to determine additional split times, in addition to the start and finish lines (e.g., start line, 10 yd [9 m], 20 yd [18 m], finish line at 40 yd [37 m]). This enables greater insight into acceleration capacities in addition to the maximal speed.

Personnel

- One timer/recorder

Procedure

1. Have the athlete warm up and dynamically stretch for several minutes.
2. Allow at least two practice runs at submaximal speed.
3. The athlete assumes a starting position using a three- or four-point stance.
4. On an auditory signal, the athlete sprints the specified distance at maximal speed.
5. The best split times of two trials are recorded to the nearest 0.1 second.
6. Allow at least 2 minutes of active recovery or rest between trials.

Note: Normative data for the 10-meter (11 yd), 20-meter (22 yd), 40-meter (44 yd), and 40-yard (37 m) sprint are presented in [table 14.33](#) near the end of the chapter.

Balance and Stability

31 BALANCE ERROR SCORING SYSTEM (BESS)

Equipment

- Foam balance pad
- Stopwatch

Personnel

- One timer/recorder

Procedure

1. The six positions of the BESS are shown in [figure 14.26](#).
2. The three stance positions are double-leg stance with feet together, single-leg stance on the nondominant foot with contralateral leg in approximately 90 degrees of flexion, and tandem stance with the dominant foot in front of the nondominant foot (128). The test is conducted on a firm surface and on a soft surface.
3. The stances are held for 20 seconds with eyes closed and hands on hips for each condition.
4. Athletes are told to keep as steady as possible, and if they lose balance, they attempt to regain their initial position as quickly as possible.
5. Errors include opening eyes; lifting the hands from the hips; touchdown of nonstance foot; step, hop, or other movement of the stance foot or feet; lifting the forefeet or heel; moving the hip into more than 30 degrees of hip flexion or abduction; or remaining out of position for more than 5 seconds.
6. The error scores from the BESS test are summed into a single score.

Note: Normative data for the BESS are presented in [table 14.34](#) near the end of the chapter.



FIGURE 14.26 Balance error scoring system (BESS): (a-c) firm surface condition and (d-f) soft surface condition.

32 STAR EXCURSION BALANCE TEST (SEBT)

Equipment

- Adhesive tape

Personnel

- One recorder

Procedure

1. The athlete stands in the center of a grid with eight lines (120 cm [47 in.]) extending out at 45-degree increments as shown in [figure 14.27](#) (115, 126).
2. The athlete maintains a single-leg stance facing in one direction while reaching with the contralateral leg as far as possible for each taped line, touching the farthest point possible and

then returning to the bilateral position. Within a single trial, the athlete remains facing in the beginning direction and the stance leg remains the same, with the other leg doing all of the reaching.

3. The distance from the center of the star to the touch position is measured.
4. The starting direction and support leg are chosen randomly. Three trials are performed for each condition and averaged.
5. A 15-second rest is allowed between each reach.
6. Trials are discarded if the athlete does not touch the line, lifts the stance foot from the center grid, loses balance, or does not maintain start and return positions for 1 full second.
7. Athletes should be given a minimum of four practice trials before being tested (103).
8. It has been suggested that testing the anteromedial, medial, and posteromedial positions is sufficient for most situations (66).

Note: Normative data for the star excursion balance test are presented in [tables 14.20](#) and [14.21](#) near the end of the chapter.

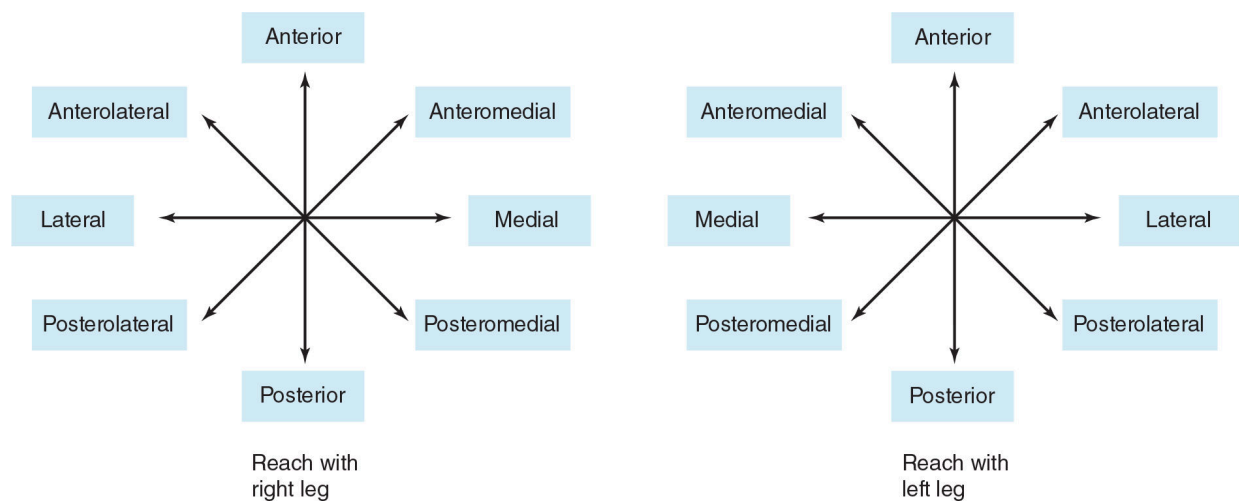


FIGURE 14.27 Directions for the star excursion balance test (SEBT).

Reprinted by permission from M.P. Reiman and R.C. Manske, *Functional Testing in Human Performance* (Human Kinetics, 2009), 109.

Flexibility

33 SIT-AND-REACH TEST

Note: A consistent method for the sit-and-reach test should be maintained if the test is done periodically. For example, if the test is performed with a measuring tape or stick during initial testing of the athlete, all subsequent testing of the athlete should be performed with a measuring tape or stick (i.e., a sit-and-reach box should not be used instead).

Equipment

- Measuring tape or stick
- Adhesive tape
- Permissible alternative: a standard sit-and-reach box

Personnel

- One tester/recorder

Procedure

1. Tape the measuring stick or tape measure to the floor. Place one piece of tape about 24 inches (61 cm) long across the measuring stick and at a right angle to it at the 15-inch (38 cm) mark.
2. Have the athlete warm up with nonballistic exercises involving the hamstrings and lower back (e.g., by walking rapidly for 3-5 min); performing several repetitions of flexing forward from a standing, knees-extended position, reaching toward the toes, then reaching upward toward the ceiling (all without jerking); jogging in place while trying to kick the heels into the upper thighs from behind; and finishing with standing toe-touching or similar stretching on the floor.
3. Have the athlete sit shoeless with the measuring stick between the legs with its zero end toward the body, the feet 12 inches (30 cm) apart, the toes pointed upward, and the heels nearly touching the edge of the taped line at the 15-inch (38 cm) mark ([figure 14.28a](#)).
4. The athlete slowly reaches forward with both hands as far as possible on the measuring stick, holding this position momentarily. To get the best stretch, the athlete should exhale and drop the head between the arms when reaching. Be sure the athlete keeps the hands adjacent to each other and does not lead with one hand. The fingertips should remain in contact with the measuring stick ([figure 14.28b](#)). The tester may hold the athlete's knees down, if necessary, to keep them straight. A score of less than 15 inches indicates that the athlete could not reach the bottom of the feet.
5. The best of three trials is recorded to the nearest 0.25 inch or 1 centimeter.

Note: Normative data for the sit-and-reach test are presented in [tables 14.22](#) through [14.25](#) near the end of the chapter.

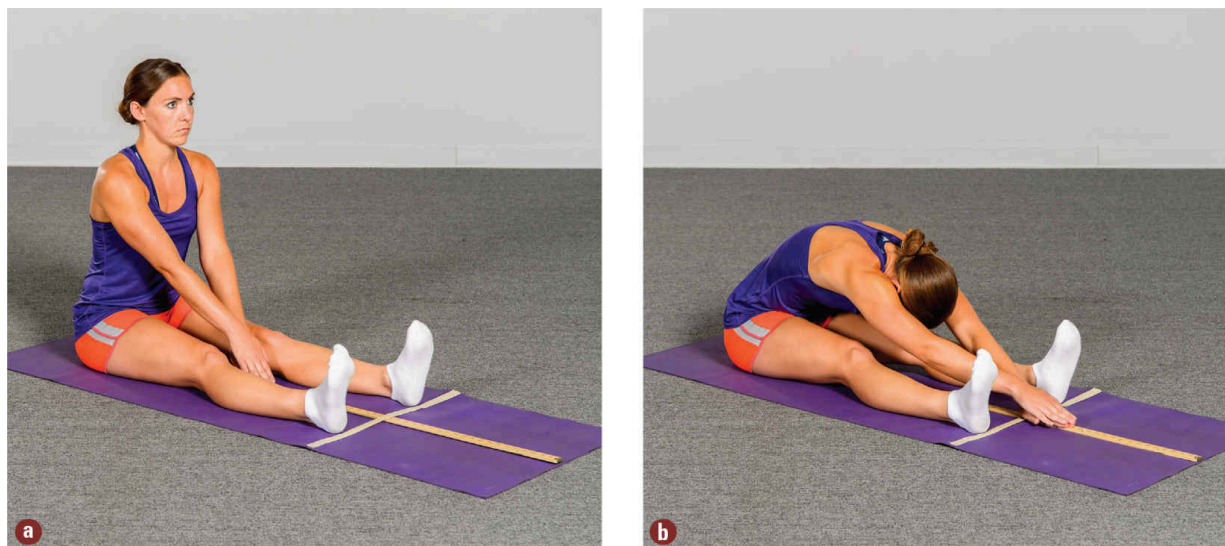


FIGURE 14.28 (a) Starting position and (b) final position of the sit-and-reach test.

34 OVERHEAD SQUAT

Equipment

- Wooden dowel or barbell

Personnel

- One tester/recorder

Procedure

1. The athlete holds the wooden dowel overhead with the shoulders fully flexed and with elbows locked. The grip should be twice shoulder-width using biacromial width as the reference measurement point for shoulder width. Biacromial width is the distance between the widest bony points at the superior portion of both shoulders. The feet are approximately shoulder-width apart, with toes pointing forward or slightly out ([figure 14.29a](#)).
2. The athlete then squats down; the initial action is flexion of the hips and knees. The heels remain in contact with the floor at all times.
3. The lowering continues until the crease of the hips is below the top of the knee.
4. The athlete should be able to hold this position with the torso remaining upright (parallel to the tibia) and the wooden dowel (or barbell) comfortably overhead ([figure 14.29b](#)).
5. The athlete performs a minimum of 5 repetitions, and the tester views the movement from the side.
6. The assessment is qualitative and the goal is to assess the movement competency, with the movement scored as pass/fail.
7. It is important that the athlete be warmed up and familiarized with the movement patterns to increase the test validity.

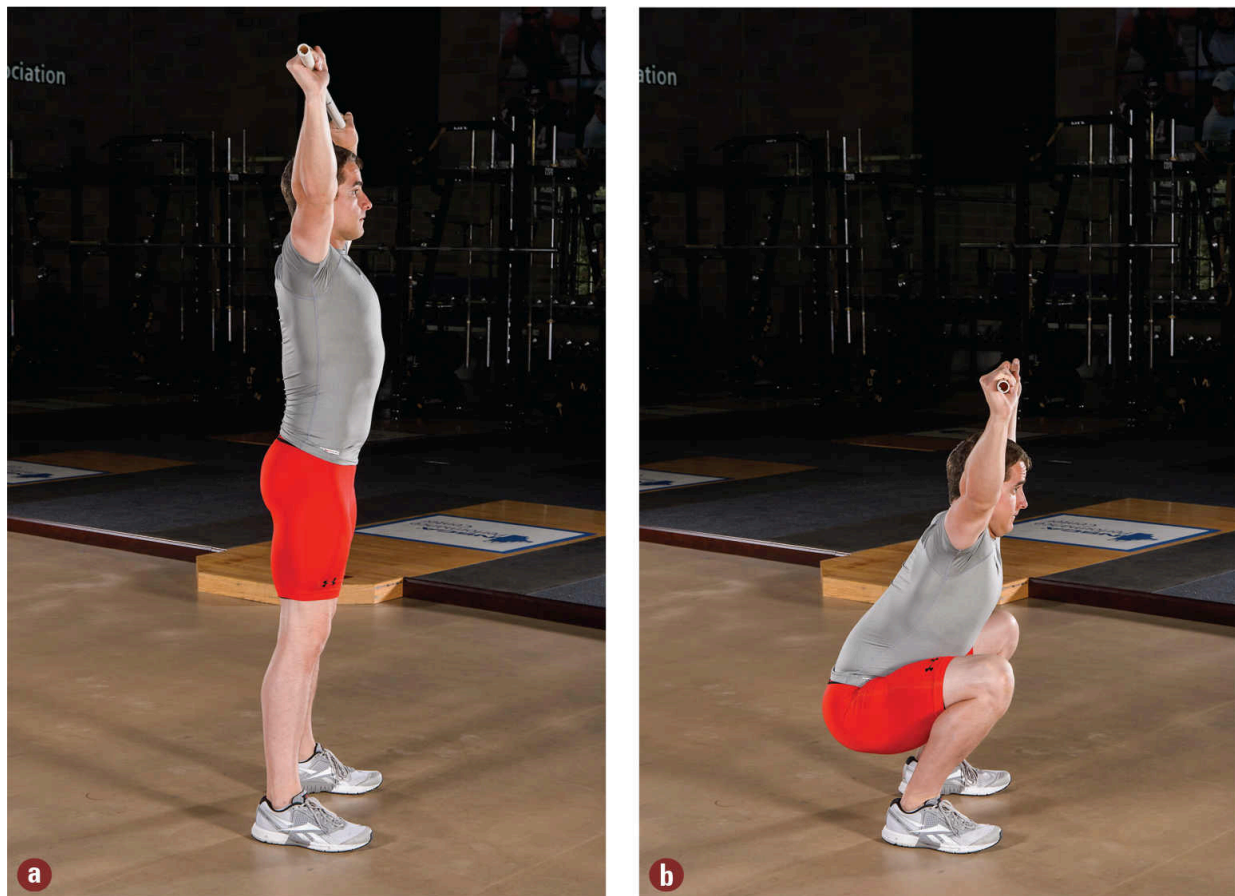


FIGURE 14.29 (a) Starting position and (b) squat position for the overhead squat.

Body Composition

35 SKINFOLD MEASUREMENTS

Equipment

- Skinfold calipers
- Flexible tape measure
- Marking pen

Personnel

- One tester, one recorder

Procedure (Obtaining a Skinfold Measurement)

1. Skinfold measurements should be made on dry skin, before exercise, to ensure maximum validity and reliability (70). The number of sites and equations should be selected based on the population tested (see [table 14.35](#) near the end of the chapter).
2. Grasp the skin firmly with the thumb and index finger to form a fold of skin and subcutaneous fat.
3. Place the caliper prongs perpendicular to the fold 0.5 inch to 1 inch (approximately 1-2 cm) from the thumb and index finger.
4. Release the caliper grip so that its spring tension is exerted on the skinfold.
5. Between 1 and 2 seconds after the grip on the caliper has been released, read the dial on the caliper to the nearest 0.5 millimeter.
6. Obtain one measurement from each test site, and then repeat all test sites for a second trial. If the measurements do not differ by more than 10%, average the two measurements to the nearest 0.5 millimeter. Otherwise, take one or more additional measurements until two of the measurements are within 10%, and average those two measurements to the nearest 0.5 millimeter.

Procedure (Measuring the Selected Site and Calculating the Body Fat Percentage)

1. There are specific equations for estimating body density (D_b) (then, in turn, percent body fat [%BF]) for different populations. First, select the equation appropriate for the athlete from [table 14.35](#) near the end of the chapter.
2. Refer to the chosen equation and related instructions and mark the skin at the appropriate anatomical sites (69, 120):
 - *Chest*—a diagonal fold one-half the distance between the anterior axillary line and the nipple for men ([figure 14.30a](#))
 - *Thigh*—a vertical fold on the anterior aspect of the thigh, midway between the hip and knee joints ([figure 14.30b](#))
 - *Abdomen*—a vertical fold 1 inch (2.5 cm) to the right (relative to the athlete) of the umbilicus ([figure 14.30c](#))
 - *Triceps*—a vertical fold on the posterior midline of the upper arm (over the triceps muscle), halfway between the acromion and the olecranon processes (the arm should be in anatomical position with the elbow extended and relaxed [[figure 14.30d](#)])
 - *Suprailium*—a diagonal fold above the crest of the ilium at the spot where an imaginary line would come down from the anterior axillary line ([figure 14.30e](#)) (some prefer the measure to be taken more laterally, at the midaxillary line)
 - *Midaxilla*—a vertical fold on the midaxillary line at the level of the xiphoid process of the sternum ([figure 14.30f](#))
 - *Subscapula*—a fold taken on a diagonal line that extends from the vertebral border to a point 0.5 inch to 1 inch (1 to 2 cm) from the inferior angle of the scapula ([figure 14.30g](#))

- *Calf*—a vertical fold along the medial side of the calf, at the level of maximum calf circumference ([figure 14.30h](#))
3. Using the appropriate population-specific equation from [table 14.35](#), calculate the estimated body density from the skinfolds (61).
 4. Enter the body density into the appropriate population-specific equation from [table 14.36](#) near the end of the chapter to calculate the percent body fat from the body density (69).
 5. Note that there are no universally accepted norms for body composition. When strength and conditioning professionals assess an athlete's body composition, they must account for a standard error of the estimate (SEE) and report a range of percentages that the athlete falls into. Note that the minimum SEE for population-specific skinfold equations is $\pm 3\%$ to $\pm 5\%$. Therefore, if a 25-year-old male athlete's body fat is measured at 24%, there is a minimum of a 6% range (21%-27%).

Note: Descriptive data for percent body fat are presented in [tables 14.22](#) through [14.25](#) and [table 14.37](#) near the end of the chapter.



FIGURE 14.30a Chest skinfold.



FIGURE 14.30b Thigh skinfold.



FIGURE 14.30c Abdomen skinfold.



FIGURE 14.30d Triceps skinfold.



FIGURE 14.30e Suprailium skinfold.



FIGURE 14.30f Midaxilla skinfold.



FIGURE 14.30g Subscapula skinfold.



FIGURE 14.30h Calf skinfold.

36 BIOELECTRICAL IMPEDANCE (BIA)

Equipment

- BIA device

Personnel

- One recorder

Procedure

1. Program the device with the athlete's sex, weight, and height.
2. Have the athlete either hold the electrodes in the hands or stand barefoot on the electrodes depending on the type of device.
3. Press the start button and wait the specified duration of time for that device (usually under 5 seconds).
4. From the measuring device, obtain and record the body composition measurements provided.
5. This means of evaluating body composition is best done under similar circumstances. To most effectively monitor body composition changes, it is recommended that testing be done first thing in the morning before the athlete has exercised, before food or water consumption, and after having voided.

Note: Descriptive data for percent body fat are presented in [tables 14.22](#) through [14.25](#) and [table 14.37](#) near the end of the chapter.

Anthropometry

37 GIRTH MEASUREMENTS

Equipment

- Flexible, spring-loaded tape measure (e.g., a Gulick tape)

Personnel

- One tester, one recorder

Procedure

1. Position the athlete in a relaxed anatomical position for each measurement (unless otherwise indicated for a particular measurement).
2. Measure the following sites (81); see [figure 14.31](#):
 - *Chest*—at nipple level in males and at maximum circumference (above the breasts) in females
 - *Right upper arm*—at the point of maximal circumference with the elbow fully extended, palm up, and arm abducted to parallel with the floor
 - *Right forearm*—at the point of maximal circumference with the elbow fully extended, palm up, and arm abducted to parallel with the floor
 - *Waist (abdomen)*—at the level of the umbilicus
 - *Hips (buttocks)*—at the maximal protrusion of the buttocks with the heels together
 - *Right thigh*—at the point of maximal circumference, usually just below the buttocks
 - *Right calf*—at the point of maximal circumference between the knee and ankle

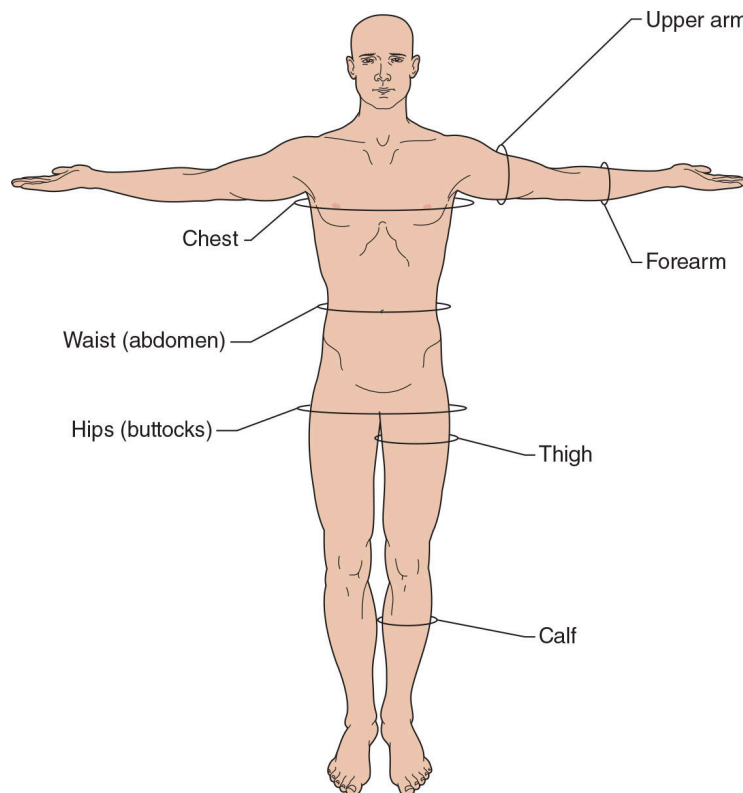


FIGURE 14.31 Sites for girth measurements.

Statistical Evaluation of Test Data

Once the proper test or tests have been chosen and administered and the scores collected, the next step may include any or all of the following: (1) analysis of the data to determine the change in performance of the individuals or group over the training period (weeks, months, or years); (2) analysis of the individual or group's performance relative to that of similar individuals or groups tested in the past; (3) analysis of the relationship of each athlete's scores to those of the group; and (4) comparison of individual scores to local, state, national, or international norms.

An important outcome of repeated performance testing is evaluation of both the improvement of individual athletes and the overall effectiveness of the physical conditioning program as determined by changes in test scores (103). A **difference score** is the difference between an athlete's score at the beginning and end of a training period or between any two separate testing times. The **percent change** is another measure that can be used. However, evaluating the effectiveness of a training program merely by degree of improvement has two major limitations. First, athletes who begin the training period at a higher training status will not improve as much as untrained athletes who perform poorly at the beginning of training (i.e., the **principle of diminishing returns**). The window of adaptation for the various physical capacities is typically greater for less well-trained athletes as well as measures of aerobic fitness (109). Second, athletes may deliberately fail to give maximal effort on pretraining tests to inflate their pre- to posttraining improvement scores. It is important to encourage athletes to give maximal effort on both the pre- and posttraining tests.

Types of Statistics

Statistics is the science of collecting, classifying, analyzing, and interpreting numerical data (27, 145). A working knowledge of statistics is helpful in making sound evaluations of test results. Within statistics are different branches such as descriptive and inferential. Over the years, scientists and practitioners in strength and conditioning have made increasing use of magnitude-based approaches (through calculation of the effect size), which can be more meaningful because they provide information regarding the magnitude of change that is practically meaningful to the athlete in a given sport.

Descriptive Statistics

Descriptive statistics summarizes or describes a large group of data. It is used when all the information about a population is known. For example, if all the members of a team are tested, statements can be made about the team with the use of descriptive statistics. There are three categories of numerical measurement in descriptive statistics: central tendency, variability, and percentile rank. In the sections that follow, these terms are defined, and examples of how to calculate the values and scores are presented.

Central Tendency Measures of **central tendency** are values about which the data tend to cluster. The three most common measures of central tendency (27, 145) are as follows:

Mean—the average of the scores (i.e., the sum of the scores divided by the number of scores). This is the most commonly used measure of central tendency.

Median—the middlemost score when a set of scores is arranged in order of magnitude. With an even number of scores, the median is the average of the two middlemost scores. Half a group of scores falls above the median and half falls below the median. Depending on the distribution of scores, the median can be a better measure of central tendency than the mean. This is particularly true when very high or very low scores of one or a few members of the group tested raise or lower the group mean to the extent that it does not adequately describe the ability of most group members.

Mode—the score that occurs with the greatest frequency. If each numerical score appears only once, there is no mode. If two or more scores are tied for greatest frequency, all of the similar scores are modes. The mode is generally regarded as the least useful measure of central tendency.

Variability The degree of dispersion of scores within a group is called **variability**. Two common measures of variability are the **range** and the **standard deviation**. The range is the interval from the lowest to the highest score. The advantage of the range is that it is easy to understand; the disadvantage is that it uses only the two extreme scores and so may not be an accurate measure of variability (145). For example, the range could be the same for a group of widely dispersed scores as for a group of scores that are narrowly dispersed except for one deviant score. The standard deviation is a measure of the variability of a set of scores about the mean. The formula for the standard deviation of a sample is as follows:

$$SD = \sqrt{\sum (x - \bar{x})^2 / n - 1} \quad (14.1)$$

where Σ refers to a summation, x is a score, $\bar{x}$ is the mean of the scores, and n is the sample size (number of scores). A relatively small standard deviation indicates that a set of scores is closely clustered about the mean; a large standard deviation indicates wider dispersion of the scores about the mean. The standard deviation is most useful when the group of scores is “normally distributed,” forming the bell-shaped curve shown in [figure 14.32](#) (26, 67).

The z-score can be used to express the distance of any individual score in standard deviation (SD) units from the mean:

$$z = (x - \bar{x}) / SD \quad (14.2)$$

For example, if an athlete runs the 40-yard (37 m) sprint in 4.6 seconds and the mean and standard deviation for the group tested are 5.00 and 0.33 seconds, respectively, equation 14.2 can be applied to determine that the z-score for that athlete is -1.2 . In other words, the athlete’s score is 1.2 standard deviation units below (i.e., faster than) the group mean. Graphs can be a useful way of representing z-scores visually. This can provide the strength and conditioning professional with a comparison of different physical capacities and provide assistance with making decisions on which weaknesses to target with a training program. In the example shown, the strength and conditioning professional may decide to focus on improving endurance and flexibility while also improving body composition ([figure 14.33](#)).

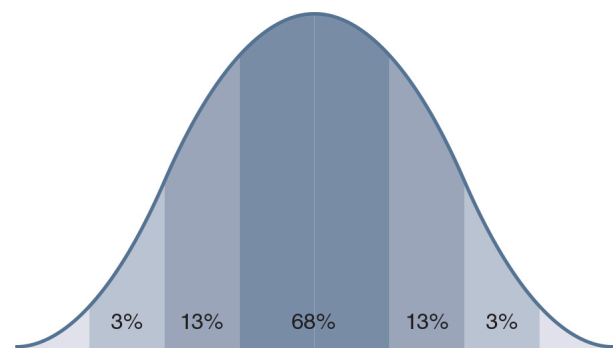


FIGURE 14.32 Normal bell curve.

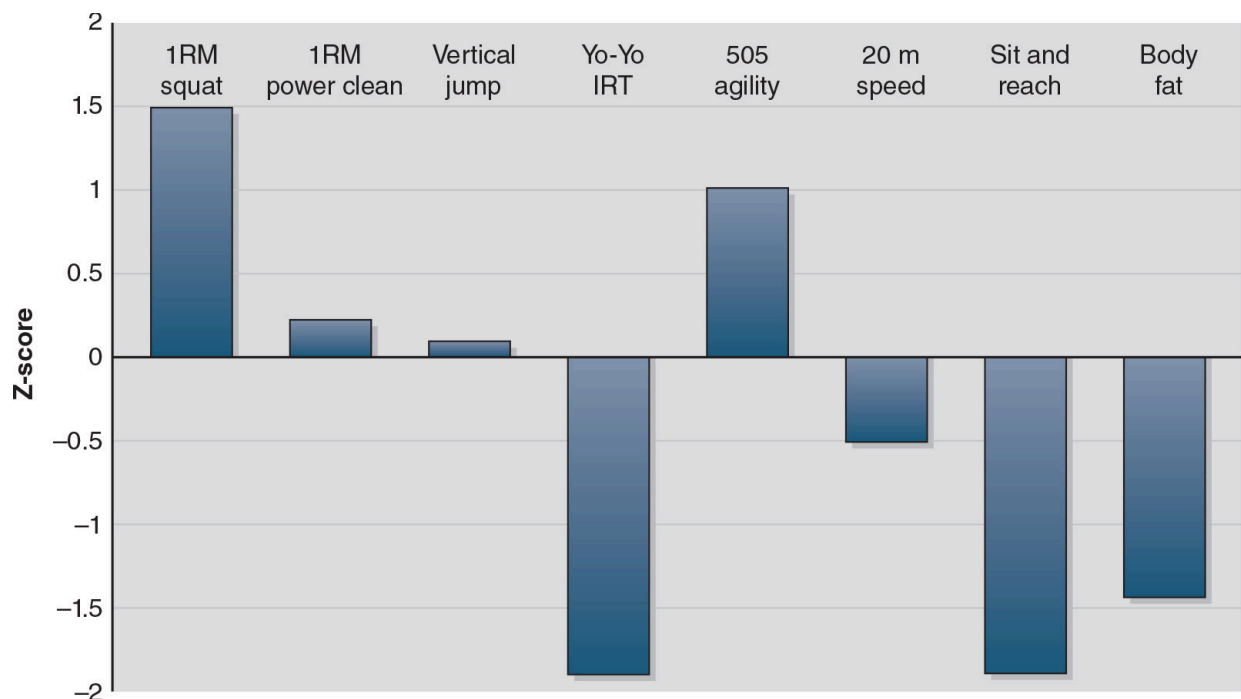


FIGURE 14.33 Z-score column chart representing standardized test scores across different fitness tests for an individual athlete. The zero represents the team average.

Percentile Rank An individual's **percentile rank** is the percentage of test takers scoring below that individual. As in calculation of the median, percentile ranking requires arranging scores ordinally (lowest to highest). For example, if an athlete is ranked in the 75th percentile, 75% of the group produced scores below that athlete's score. Norms based on large samples are sometimes expressed in evenly spaced percentiles. Several examples of percentile rank tables are shown in the tables near the end of the chapter.

Inferential Versus Magnitude Statistics

The use of **inferential statistics** allows one to draw general conclusions about a population from information collected in a population sample. For example, if a boys' 9th-grade gym class is put through a battery of tests and it is assumed that the class (sample) is representative of all the 9th-grade boys in the school (the population), the results of these tests can be used to make inferences about the population as a whole. A basic assumption of inferential statistics is that the sample is truly representative of the population (27).

Magnitude statistics can provide a more useful approach for practitioners because it allows for interpretation of the clinical significance of fitness testing (75). To describe and evaluate the magnitude of change in a fitness test, measures such as smallest worthwhile change and effect size are important.

Smallest worthwhile change refers to the ability of a test to detect the smallest practically important change in performance. The ability to track changes with a fitness test depends on the validity and reliability of that test. The smallest worthwhile change can be determined in a number of ways, but it is typically calculated as 0.2 of the between-subjects standard deviation (75). For example, if the standard deviation for a vertical jump test is 10 centimeters in a group of female athletes, this would mean that the smallest worthwhile change for this group of athletes is 2 centimeters (0.2×10 cm).

Effect size is a statistic that can be useful for calculating group performance following a training program or comparing between groups of athletes (42). The effect of the training program can be calculated as the difference or change in the mean score as a proportion of the pretest standard deviation (equation 14.3).

$$ES = (x_{\text{posttest}} - x_{\text{pretest}}) / SD_{\text{pretest}} \quad (14.3)$$

For example, a group of athletes had a pretraining mean bench press 1RM of 104.5 kilograms (standard deviation of 5.7 kg), and after a 12-week training intervention the mean bench press is 111.7 kilograms. The calculated effect size would be equal to $(111.7 - 104.5) / 5.7 = 1.26$.

Several scales have been provided to compare the magnitude of the effect (28, 103, 127), but reference values for small (0.2), moderate (0.6), large (1.2), and very large (2.0) can be a useful starting point for practitioners (42, 75). For the example just given, this would mean that the strength and conditioning professional would interpret the effect size of 1.26 as meaning that the training program had a large effect.

Developing an Athletic Profile

To determine the sport-specific training status of an athlete, the strength and conditioning professional can combine the results of selected tests to generate an **athletic profile**, which is a group of test results related to sport-specific abilities that are important for quality performance in a sport or sport position. When evaluating athletes, the strength and conditioning professional should follow these six steps:

1. Select tests that will measure the specific parameters most closely related to the physical characteristics of the sport or sports in question. For example, a testing battery for wrestlers should include tests for pulling strength, pushing strength, and local muscular endurance.
2. Choose valid and reliable tests to measure these parameters, and arrange the testing battery in an appropriate order with sufficient rest between tests to promote test reliability (see [chapter 13](#) for more detail). For example, appropriate tests for wrestling might include push-ups and sit-ups for maximum repetitions in a given time interval. These two tests should be separated by at least 10 minutes of rest to allow recovery from fatigue and thus promote accurate scores.
3. Administer the test battery with as many athletes as possible.
4. Determine the smallest worthwhile change for the tests and compare to normative data where appropriate. It is recommended that coaches store testing results and develop their own norms when standardized procedures are used.
5. Conduct repeat testing (e.g., pre- and posttraining program) and use the results to present a visual profile with figures.
6. Use the results of the testing in some meaningful way. Ideally, the results will enable the strength and conditioning professional to identify the strengths and weaknesses of the athletes and to design the training program with these in mind.

Conclusion

Motor abilities and body composition variables that can be improved through strength and conditioning programs include maximum muscular strength, maximum muscular power, anaerobic capacity, local muscular endurance, aerobic capacity, agility, speed, flexibility, girths, percent body fat, and lean body mass. Performance testing can be used to evaluate basic motor abilities as well as the improvement of individual athletes over time and the overall effectiveness of a physical conditioning program. Numerous tests are available to measure sport-specific physical capabilities and training status. Strength and conditioning professionals can either use existing normative data to evaluate athletic performance or develop their own normative data. Statistical measures of central tendency, variability, percentile rank, smallest worthwhile change, effect size, and standard scores are useful for evaluating physical abilities and the improvement of a group as well as the individuals within the group.

KEY TERMS

- aerobic capacity
- aerobic power
- agility
- anaerobic capacity
- anaerobic power
- anthropometry
- artificial intelligence (AI)
- athletic motor skill competencies (AMSC)
- athletic performance
- athletic profile
- balance
- body composition
- central tendency
- descriptive statistics
- difference score
- effect size
- flexibility
- functional movement screen (FMS)
- high-speed muscular strength
- inferential statistics
- local muscular endurance
- low-speed muscular strength
- machine learning (ML)
- magnitude statistics
- maximal anaerobic muscular power
- mean
- median
- mode
- percent change
- percentile rank
- principle of diminishing returns
- procedures
- protocols

range
 reactive strength index
 smallest worthwhile change
 speed
 stability
 standard deviation
 statistics
 variability

STUDY QUESTIONS

- The maximal rate of energy production by the combined phosphagen and glycolytic energy system is known as
 - agility
 - anaerobic capacity
 - aerobic capacity
 - maximal strength
- One of the following tests is not a test of agility or change of direction.
 - hexagon test
 - 5-10-5 (pro agility) test
 - T-test
 - reactive strength index
- Which of the following is a disqualification for the Nordic hamstring test?
 - allowing hip flexion rather than maintaining a straight line from the knees to the head
 - going too slow
 - allowing the ankles alone to contact the hooks when the athlete is using a dynamometer
 - maintaining tight core control
- With an even number of scores, the _____ is the average of the two middlemost scores. Half a group of scores falls above this, and half falls below.
 - median
 - mode
 - average
 - variability
- Which of the following tests is a measurement of anthropometry?
 - girth measurements
 - body composition
 - isometric mid-thigh pull
 - 1RM power clean

TABLE 14.1 Percentile Values of the 1RM Bench Press, Squat, and Power Clean in NCAA Division I Female Collegiate Athletes

	1RM bench press		1RM squat		1RM power clean		1RM bench press		1RM squat	
	lb	kg	lb	kg	lb	kg	lb	kg	lb	kg
Percentile*	BASKETBALL						SWIMMING			

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

lb = pounds, SD = standard deviation, n = sample size

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

90	124	56	178	81	130	59	116	53	145	66
80	119	54	160	73	124	56	109	50	135	61
70	115	52	147	67	117	53	106	48	129	59
60	112	51	135	61	112	51	101	46	120	55
50	106	48	129	59	110	50	97	44	116	53
40	102	46	115	52	103	47	94	43	112	51
30	96	44	112	51	96	44	93	42	104	47
20	88	40	101	46	88	40	88	40	101	46
10	82	37	81	37	77	35	78	35	97	44
Mean	105	48	130	59	106	48	98	45	118	54
SD	18	8	42	19	20	9	15	7	19	9
<i>n</i>	120		86		85		42		35	
Percentile*	SOFTBALL						VOLLEYBALL			
90	117	53	184	84	122	55	113	51	185	84
80	108	49	170	77	115	52	108	49	171	78
70	104	47	148	67	106	48	104	47	165	75
60	99	45	139	63	100	45	100	45	153	70
50	95	43	126	57	94	43	98	45	143	65
40	90	41	120	55	93	42	96	44	136	62
30	85	39	112	51	88	40	90	41	126	57
20	80	36	94	43	80	36	85	39	112	51
10	69	31	76	35	71	32	79	36	98	45
Mean	94	43	130	59	97	44	97	44	144	65
SD	18	8	42	19	20	9	14	6	33	15
<i>n</i>	105		97		80		67		62	

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

lb = pounds, SD = standard deviation, *n* = sample size

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

TABLE 14.2 Percentile Values of the 1RM Bench Press, Squat, and Power Clean in High School and College American Football Players

Percentile*	1RM bench press		1RM squat		1RM power clean		1RM bench press		1RM squat		1RM power clean	
	lb	kg	lb	kg	lb	kg	lb	kg	lb	kg	lb	kg
	HIGH SCHOOL 14-15 YEARS						HIGH SCHOOL 16-18 YEARS					
90	243	110	385	175	213	97	275	125	465	211	250	114

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

**Hoffman 2006 reported 365 lb (166 kg) for the 40% rank of the NCAA DIII 1RM squat.

lb = pounds, SD = standard deviation, *n* = sample size.

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

80	210	95	344	156	195	89	250	114	425	193	235	107
70	195	89	325	148	190	86	235	107	405	184	225	102
60	185	84	305	139	183	83	225	102	365	166	223	101
50	170	77	295	134	173	79	215	98	335	152	208	95
40	165	75	275	125	165	75	205	93	315	143	200	91
30	155	70	255	116	161	73	195	89	295	134	183	83
20	145	66	236	107	153	70	175	80	275	125	165	75
10	125	57	205	93	141	64	160	73	250	114	145	66
Mean	179	81	294	134	176	80	214	97	348	158	204	93
SD	45	20	73	33	32	15	44	20	88	40	43	20
<i>n</i>	214		170		180		339		249		284	
Percentile*	NCAA DI						NCAA DIII					
90	370	168	500	227	300	136	365	166	470	214		
80	345	157	455	207	280	127	325	148	425	193		
70	325	148	430	195	270	123	307	140	405	184		
60	315	143	405	184	261	119	295	134	385	175		
50	300	136	395	180	252	115	280	127	365	166		
40	285	130	375	170	242	110	273	124	350**	159**		
30	270	123	355	161	232	105	255	116	335	152		
20	255	116	330	150	220	100	245	111	315	143		
10	240	109	300	136	205	93	225	102	283	129		
Mean	301	137	395	180	252	115	287	130	375	170		
SD	53	24	77	35	38	17	57	26	75	34		
<i>n</i>	1,189		1,074		1,017		591		588			

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

**Hoffman 2006 reported 365 lb (166 kg) for the 40% rank of the NCAA DIII 1RM squat.

lb = pounds, SD = standard deviation, *n* = sample size.

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

TABLE 14.3 Percentile Values of the 1RM Bench Press, Squat, and Power Clean in NCAA Division I Male Baseball and Basketball Athletes

	1RM bench press		1RM squat		1RM power clean		1RM bench press		1RM squat		1RM power clean	
	lb	kg	lb	kg	lb	kg	lb	kg	lb	kg	lb	kg
Percentile*	BASEBALL						BASKETBALL					
90	273	124	365	166	265	120	269	122	315	143	250	114

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

lb = pounds, SD = standard deviation, *n* = sample size.

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

80	260	118	324	147	239	109	250	114	305	139	235	107
70	247	112	310	141	225	102	240	109	295	134	230	105
60	239	109	293	133	216	98	230	105	280	127	220	100
50	225	102	270	123	206	94	225	102	265	120	215	98
40	218	99	265	120	200	91	216	98	245	111	205	93
30	203	92	247	112	190	86	210	95	225	102	195	89
20	194	88	237	107	182	83	195	89	195	89	180	82
10	175	80	218	99	162	74	185	84	166	75	162	74
Mean	227	103	281	128	210	95	225	102	251	114	209	95
SD	41	19	57	26	36	16	33	15	57	26	34	15
<i>n</i>	170		176		149		142		131		122	

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

lb = pounds, SD = standard deviation, *n* = sample size.

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

**TABLE 14.4 1RM Squat, Bench Press, Power Clean, and Bench Pull
Descriptive Data* for Various Groups**

Group, sport, or position	Number of athletes	Body weight		Squat		Bench press		Power clean		Bench pull	
		lb	kg	lb	kg	lb	kg	lb	kg	lb	kg
Group, sport, or position	Number of athletes	Body weight		Squat		Bench press		Power clean		Bench pull	
		lb	kg	lb	kg	lb	kg	lb	kg	lb	kg
National rugby league (7)	20	216.5 ± 21.8	98.2 ± 9.9	385.8 ± 59.5	175.0 ± 27.3						
State-level rugby league (7)	20	201.2 ± 18.5	91.3 ± 8.4	329.8 ± 30.9	149.6 ± 14.3						
National rugby league (8)	21	219.4 ± 19.2	99.5 ± 8.7			315.0 ± 33.5	142.7 ± 15.2				
State-level rugby league (8)	21	200.0 ± 19.6	90.7 ± 8.9			258.8 ± 35.9	117.4 ± 16.3				
National rugby league (5)	6	211.2 ± 32.2	95.8 ± 14.6			332.5 ± 2.6	150.8 ± 10.7				
National sailing—men (119)	11	215.6 ± 27.6	97.8 ± 12.5			310.6 ± 58.6	140.9 ± 26.6			265.9 ± 37.3	120.6 ± 16.9
National canoe/rowing/wrestling/judo—men (129)	75	167.6 ± 19.4	76.0 ± 8.8			199.0 ± 35.9	90.3 ± 16.3			176.8 ± 26.0	80.2 ± 11.8
National basketball—women (140)	12	166.7 ± 32.2	75.6 ± 14.6			173.2 ± 6.6	78.6 ± 3.0				
NCAA Division IA college football—men (17)	963					310.4	140.8				
NCAA Division IA college football—men (17)	560			420.6	190.8						

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

National softball—women (113)	10	158.1 ± 24.5	71.7 ± 11.1	183.6 ± 22.0	83.3 ± 10.0						
National ice hockey—women (123)	22	155.2 ± 15.7	70.4 ± 7.1			144.0 ± 27.0	65.3 ± 12.2				
National soccer—men (149)	17	168.7 ± 16.8	76.5 ± 7.6	378.5 ± 46.7	171.7 ± 21.2						
National soccer—men (161)	14	169.5 ± 13.9	76.9 ± 6.3	362.9 ± 43.9	164.6 ± 21.8	182.3 ± 28.2	82.7 ± 12.8				
National soccer—men (161)	15	169.6 ± 16.3	76.8 ± 7.4	297.6 ± 35.7	135.0 ± 16.2	170.0 ± 36.4	77.1 ± 16.5				
NCAA Division IA college football—men (114)	207	229.2 ± 48.7	104.0 ± 22.1			312.8 ± 58.6	141.6 ± 26.6				
NCAA Division IA college football—men (114)	88	226.0 ± 14.3	102.5 ± 6.5					267.4 ± 33.1	121.3 ± 15.0		
NCAA Division IA college football—men (114)	86	226.0 ± 14.3	102.5 ± 6.5	397.9 ± 74.1	180.5 ± 33.6						
National handball—women (55)	16	152.7 ± 18.1	69.3 ± 8.2			113.8 ± 14.8	51.6 ± 6.7				
National handball—men (54)	15	207.0 ± 37.3	93.9 ± 16.9			235.7 ± 25.6	106.9 ± 11.6				
Under 16 rugby league—men (149)	30	165.8 ± 24.5	75.2 ± 11.1	221.3 ± 48.3	100.4 ± 21.9	162.9 ± 29.1	73.9 ± 13.2			156.3 ± 22.3	70.9 ± 10.1
Under 17 rugby league—men (149)	48	178.8 ± 20.7	81.1 ± 9.4	269.4 ± 41.2	122.2 ± 18.7	205.7 ± 29.5	93.3 ± 13.4			184.1 ± 22.5	83.5 ± 10.2
Under 18 rugby league—men (149)	55	188.1 ± 22.0	85.3 ± 10.0	295.4 ± 34.2	134.0 ± 15.5	228.6 ± 33.7	103.7 ± 15.3			200.8 ± 22.3	91.1 ± 10.1
Under 19 rugby league—men (149)	45	195.8 ± 21.8	88.8 ± 9.9	305.1 ± 43.2	138.4 ± 19.6	249.8 ± 36.2	113.3 ± 16.4			215.2 ± 27.3	97.6 ± 12.4
Under 20 rugby league—men (149)	26	196.0 ± 18.7	88.9 ± 8.5	318.8 ± 48.7	144.6 ± 22.1	252.0 ± 33.7	114.3 ± 15.3			220.5 ± 24.7	100.0 ± 11.2
NCAA Division III lacrosse—women (73)	11	132.3 ± 11.0	60.0 ± 5.0	158.5 ± 13.4	71.9 ± 6.1	96.8 ± 13.4	43.9 ± 6.1				
National rugby league forwards—men (102)	63					271.2 ± 26.0	123.0 ± 11.8				
National rugby league backs—men (102)	55					251.3 ± 37.5	114.0 ± 17.0				
National rugby union—men (30)	30	236.1 ± 22.3	107.1 ± 10.1	351.6 ± 58.0	159.5 ± 26.3	308.6 ± 35.9	140.0 ± 16.3				
National rugby union—women (11)	15	157.9 ± 21.8	71.6 ± 9.9	152.1 ± 19.4	69.0 ± 8.8						

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

TABLE 14.5 Descriptive Data* for Bench Press, Back Squat, Vertical Jump, and Standing Long Jump in Female NCAA Division III Athletes

Sport	Bench press (kg)	Back squat (kg)	Vertical jump (cm)	Standing long jump (cm)
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*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

Adapted by permission from B. Guthrie, J.B. Fields, B. Thompson, and M.T. Jone, "Physical Performance Assessments of Strength and Power in Women Collegiate Athletes," *International Journal of Exercise Science* 14, no. 6 (2021): 984-993.

Field hockey	43.9 ± 7.4	78.4 ± 16.44	44.1 ± 6.5	174.4 ± 16.5
Softball	46.1 ± 9.5	83.8 ± 17.9	52.5 ± 4.5	191.4 ± 11.9
Soccer	42.9 ± 7.9	72.5 ± 13.9	46.2 ± 6.3	178.1 ± 16.5
Volleyball	42.7 ± 7.5	76.1 ± 14.0	50.8 ± 7.2	N/A

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

Adapted by permission from B. Guthrie, J.B. Fields, B. Thompson, and M.T. Jone, "Physical Performance Assessments of Strength and Power in Women Collegiate Athletes," *International Journal of Exercise Science* 14, no. 6 (2021): 984-993.

TABLE 14.6 Descriptive Percentile Values for Isometric Mid-Thigh Pull Peak Force, Relative Peak Force, and Rate of Force Development for Male Senior and Junior Soccer Players of National and Regional Standard

Percentile	Age group	Level of competition	Peak force	Relative peak force	Rate of force development (0-250 ms)
			(N)	(N/kg)	(N/s)
100	Senior	National (<i>n</i> = 25)	3,698	46.7	6,814
		Regional (<i>n</i> = 24)	3,702	49.8	7,237
	Junior	National (<i>n</i> = 32)	3,714	51.4	6,387
		Regional (<i>n</i> = 30)	3,296	42.1	6,603
75	Senior	National (<i>n</i> = 25)	3,089	42.1	5,588
		Regional (<i>n</i> = 24)	2,995	42.4	5,113
	Junior	National (<i>n</i> = 32)	2,965	41.0	5,457
		Regional (<i>n</i> = 30)	2,749	38.7	4,466
50	Senior	National (<i>n</i> = 25)	2,917	38.6	5,171
		Regional (<i>n</i> = 24)	2,595	38.9	4,005
	Junior	National (<i>n</i> = 32)	2,737	39.4	4,676
		Regional (<i>n</i> = 30)	2,347	35.9	3,902
25	Senior	National (<i>n</i> = 25)	2,725	36.8	4,876
		Regional (<i>n</i> = 24)	2,486	35.0	3,573
	Junior	National (<i>n</i> = 32)	2,504	38.0	4,394
		Regional (<i>n</i> = 30)	2,153	33.1	3,534
0	Senior	National (<i>n</i> = 25)	2,262	33.0	2,221
		Regional (<i>n</i> = 24)	2,264	28.1	2,429
	Junior	National (<i>n</i> = 32)	2,284	35.7	2,721
		Regional (<i>n</i> = 30)	1,960	30.8	1,987

Reprinted by permission from M.A. Soriano, V. Paredes, P. Comfort, et al., "'You Are Not Wrong About Getting Strong:' An Insight Into the Impact of Age Group and Level of Competition on Strength in Spanish Football Players," *International Journal of Sports Physiology and Performance* 19, no. 7 (2024): 633.

TABLE 14.7 Normative Data for Nordic Hamstring Test

Group	Absolute maximal eccentric Nordic strength (Mean $\pm$ SD, <i>n</i>)	Relative maximal eccentric Nordic strength (Mean $\pm$ SD, <i>n</i> · kg ⁻¹)
FOOTBALL (<i>n</i> = 174)	276.29 $\pm$ 74.76	4.95 $\pm$ 0.95
ATHLETICS (<i>n</i> = 123)	289.73 $\pm$ 94.60	4.62 $\pm$ 1.01
Sprints (<i>n</i> = 27)	347.00 $\pm$ 83.54	5.07 $\pm$ 1.10
Endurance (<i>n</i> = 40)	283.85 $\pm$ 75.20	4.90 $\pm$ 0.86
Jumps (<i>n</i> = 15)	320.17 $\pm$ 74.36	4.58 $\pm$ 0.87
Throws (<i>n</i> = 5)	417.70 $\pm$ 83.92	4.14 $\pm$ 0.85
Development (<i>n</i> = 28)	234.93 $\pm$ 74.16	4.18 $\pm$ 0.91
Foundation (<i>n</i> = 8)	180.69 $\pm$ 98.19	3.58 $\pm$ 0.86
MULTI-SPORTS (<i>n</i> = 40)	240.74 $\pm$ 77.64	4.30 $\pm$ 1.21
Fencing (<i>n</i> = 16)	226.42 $\pm$ 71.54	4.13 $\pm$ 1.16
Gymnastics (<i>n</i> = 6)	209.18 $\pm$ 57.05	3.69 $\pm$ 0.98
Squash (<i>n</i> = 12)	303.46 $\pm$ 75.22	5.17 $\pm$ 1.12
Table tennis (<i>n</i> = 6)	189.71 $\pm$ 41.07	3.58 $\pm$ 1.02
TOTAL (<i>n</i> = 337)	276.97 $\pm$ 83.92	4.74 $\pm$ 1.03

Reprinted by permission from E. Jeanguyot, B. Salcinovic, A. Johnson, et al., "Eccentric Hamstring Strength in Young Athletes Is Best Documented When Normalised to Body Mass: A Cross-Sectional Study With Normative Data of 590 Athletes From Different Age Categories," *Biology of Sport* 40, no. 4 (2023): 1079-1095. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

TABLE 14.8 Norms for Standing Long Jump in Elite Male and Female Athletes

Percentile*	Males		Females	
	in.	cm	in.	cm
90	133	339	115	293
80	122	309	110	279
70	116	294	104	264
60	110	279	98	249
50	104	264	92	234
40	98	249	86	219
30	92	234	80	204
20	86	219	74	189
10	80	204	69	174

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

Reprinted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 58. Adapted from D.A. Chu, *Explosive Power and Strength* (Human Kinetics, 1996).

TABLE 14.9 Rankings for Standing Long Jump in 15- and 16-Year-Old Male and Female Athletes

Category	Males	Females
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	in.	cm	in.	cm
Excellent	79	201	65	166
Above average	73	186	61	156
Average	69	176	57	146
Below average	65	165	53	135
Poor	<65	<165	<53	<135

Reprinted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 58. Adapted from D.A. Chu, *Explosive Power and Strength* (Human Kinetics, 1996).

TABLE 14.10 Vertical Jump, Static Jump, and Broad Jump Descriptive Data* for Various Groups

Group, sport, or position	Number of athletes	Vertical jump		Static jump		Broad jump	
		in.	cm	in.	cm	in.	cm
College soccer—women (154)	51	16.1 ± 2.2	40.9 ± 5.6				
High school soccer—women (154)	83	15.6 ± 1.9	39.6 ± 4.8				
College lacrosse—women (143)	79	15.8 ± 2.2	40.1 ± 5.6				
Under 18 Gaelic football—men (34)	265	17.0 ± 2.0	43.1 ± 5.1			78.0 ± 8.1	198.1 ± 20.6
National soccer—women (26)	21	12.4 ± 1.6	31.5 ± 4.0	11.9 ± 1.5	30.2 ± 3.8		
Under 19 soccer—women (26)	20	13.5 ± 1.5	34.3 ± 3.8	12.9 ± 1.1	32.8 ± 2.8		
Under 17 soccer—women (26)	21	11.4 ± 0.8	29.0 ± 2.0	11.1 ± 1.0	28.2 ± 2.5		
Under 21 soccer—men (26)	18	15.9 ± 1.7	40.4 ± 4.3	14.6 ± 1.5	37.1 ± 3.8		
Under 20 soccer—men (26)	17	15.8 ± 1.9	40.1 ± 4.8	15.0 ± 1.9	38.1 ± 4.8		
Under 17 soccer—men (26)	21	16.1 ± 2.0	40.9 ± 5.1	14.7 ± 1.9	37.3 ± 4.8		
Division I Spain soccer—women (132)	100	10.3 ± 1.9	26.2 ± 4.8				
National ice hockey—women (123)	23	19.8 ± 2.2	50.3 ± 5.6#			84.6 ± 4.3	214.9 ± 10.9
National soccer—women (65)	85	12.1 ± 1.6	30.7 ± 4.1				
Division I Norway soccer—women (65)	47	11.1 ± 1.6	28.2 ± 4.1				
Ice hockey National Hockey League draftees—men (24)	853	24.4 ± 3.0	62.0 ± 7.6#			100.0 ± 7.0	254.0 ± 17.8
College wrestling—men (144)	20	20.5 ± 3.1	52.1 ± 7.9#				

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

#Jumps performed with an arm swing.

National weightlifting—men (46)	6	23.9 ± 1.5	60.7 ± 3.8				
National soccer—men (160)	17	22.2 ± 1.6	56.4 ± 4.1				
National soccer—men (161)	14	22.3 ± 2.6	56.6 ± 6.6				
National soccer—men (161)	15	20.9 ± 1.6	53.1 ± 4.1				
National soccer—men (141)	270	17.8 ± 0.7	45.2 ± 1.8	17.4 ± 0.5	44.2 ± 1.3		
National handball—women (55)	16	15.1 ± 1.7	38.4 ± 4.3				
National handball—men (53)	15	19.0 ± 2.8	48.3 ± 7.1				
Under 16 rugby league—men (149)	67	18.0 ± 2.0	45.7 ± 5.1				
NCAA Division I football—men (45)	193	28.7 ± 3.7	72.9 ± 9.4#				
NCAA Division II football—men (45)	181	27.3 ± 3.3	69.3 ± 8.4#				
NCAA Division III football—men (45)	131	26.5 ± 3.5	67.3 ± 8.9#				
Under 20 rugby league—men (149)	25	20.8 ± 2.1	52.8 ± 5.3				
High school volleyball—women (131)	27	18.5 ± 3.3	47.0 ± 8.4#				
NCAA Division I volleyball—women (131)	26	20.8 ± 2.5	52.8 ± 6.4#				
Professional rugby league (148)	70	16.2 ± 2.8	41.1 ± 7.1				
Amateur rugby league (148)	249	15.4 ± 1.9	39.1 ± 4.8				
Academy rugby league (148)	261	15.7 ± 1.9	39.9 ± 4.8				
National rugby league forwards—men (30)	12	14.7 ± 1.7	37.3 ± 4.3				
National rugby league backs—men (30)	6	15.9 ± 2.5	40.4 ± 6.4				
National rugby league—men (49)	26	20.0 ± 2.9	50.8 ± 7.4#				
National rugby league—men (48)	58	24.7 ± 2.2	62.7 ± 5.6#				
National rugby union—men (33)	30					101.6 ± 7.9	258.1 ± 20.1
National rugby union—women (11)	15	15.0 ± 1.6	38.1 ± 4.1	13.8 ± 1.2	35.1 ± 3.0		

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

#Jumps performed with an arm swing.

High school track and field—women (107)	8					83.4 ± 6.3	211.8 ± 16.0
NCAA Division I soccer—women (97)	15	12.2 ± 2.0	31.0 ± 5.1			57.9 ± 4.3	147.1 ± 10.9
High school rugby league—men (147)	302	16.3 ± 2.1	41.4 ± 5.3				
Junior national volleyball—men (47)	14	21.5 ± 0.9	54.6 ± 2.3#				
Junior national volleyball—women (47)	15	18.0 ± 0.6	45.7 ± 1.5#				
Under 18 Australian rules football—men (164)	177	23.9 ± 2.2	60.7 ± 5.6#				
NCAA Division I lacrosse—women (142)	84	15.8 ± 2.2	40.1 ± 5.6				
NCAA Division I soccer—men (136)	27	24.3 ± 2.8	61.7 ± 7.1#				
National soccer—women (1)	17	12.0 ± 0.5	30.5 ± 1.3				
National soccer—women (108)	17	12.8 ± 1.5	32.5 ± 3.8				
National soccer—men (108)	17	17.2 ± 0.9	43.7 ± 2.3				
National junior soccer—women (108)	17	11.2 ± 0.8	28.4 ± 2.0				
National junior soccer—men (108)	17	17.3 ± 1.9	43.9 ± 4.8				
National soccer—men (2)	214	15.4 ± 2.0	39.1 ± 5.1	14.8 ± 1.9	37.6 ± 4.8		
NCAA Division I college basketball—men (121)	16	17.1 ± 0.4	43.4 ± 1.9				
NCAA Division I college basketball forwards—men (16)	37	16.2 ± 2.2	41.1 ± 5.6				
NCAA Division I college basketball guards—men (16)	55	17.2 ± 2.4	43.7 ± 6.1				

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

#Jumps performed with an arm swing.

TABLE 14.11 Rebound Jump Descriptive Data* for Various Groups

Population	Jump height (cm)	RSI avg	RSI max	Device
NCAA Division I college basketball forwards—men	31.1 ± 1.5^	1.40 ± 0.08^	1.47 ± 0.08	FP
Intercounty hurlers—men	31.3 ± 4.1	1.72 ± 0.34	—	OPTO
Intercounty Gaelic football players—women	26.3 ± 4.3	1.47 ± 0.28	—	OPTO
Club Gaelic football players—women	23.1 ± 4.3	1.22 ± 0.27	—	OPTO

RSI = reactive strength index; Avg = average; FP = force plate; OPTO = optoelectronic device. ^ = average values taken from the top 3 best repetitions rather than the top 5.

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

Adapted from Brady et al. (2024); Hughes et al. (2024); Parchmann and McBride (2011).

TABLE 14.12 Margaria-Kalamen Stair Sprint Test Guidelines (Watts)

Category	Age groups (years)				
	15-20	20-30	30-40	40-50	Over 50
MEN					
Excellent	Over 2,197	Over 2,059	Over 1,648	Over 1,226	Over 961
Good	1,844-2,197	1,726-2,059	1,383-1,648	1,040-1,226	814-961
Average	1,471-1,824	1,373-1,716	1,098-1,373	834-1,030	647-804
Fair	1,108-1,461	1,040-1,363	834-1,088	637-824	490-637
Poor	Under 1,108	Under 1,040	Under 834	Under 637	Under 490
WOMEN					
Excellent	Over 1,785	Over 1,648	Over 1,226	Over 961	Over 736
Good	1,491-1,785	1,383-1,648	1,040-1,226	814-961	608-736
Average	1,187-1,481	1,098-1,373	834-1,030	647-804	481-598
Fair	902-1,177	834-1,089	637-824	490-637	373-471
Poor	Under 902	Under 834	Under 637	Under 490	Under 373

Adapted by permission from E. Fox, R. Bowers, and M. Foss, *The Physiological Basis for Exercise and Sport*, 5th ed. (Brown & Benchmark, 1993), 676; permission conveyed through Copyright Clearance Center, Inc.

TABLE 14.13 Velocity-Based Training Normative Data for Barbell Velocity

	Absolute strength	Accelerative strength	Strength-speed	Speed-strength	Starting strength
% 1RM	80%-100%	60%-80%	40%-60%	20%-40%	Body weight to 20%
Lower body speed range	<0.50 m/s	0.50-0.75 m/s	0.75-1.0 m/s	1.0-1.3 m/s	>1.3 m/s
Upper body speed range	<0.40 m/s	0.40-0.60 m/s	0.60-0.85 m/s	0.85-1.1 m/s	>1.1 m/s

Reprinted by permission from J.B. Mann, *Developing Explosive Athletes: Use of Velocity Based Training in Training Athletes*, 4th ed. (Ultimate Athlete Concepts, 2021).

TABLE 14.14 Descriptive Data* for 300-Yard (274 m) Shuttle Test

Group, sport, or position	Number of athletes	Time (s)
High school volleyball—women (131)	27	68.0 ± 6.3
NCAA Division 1 volleyball—women (131)	26	67.7 ± 3.8
National soccer—men (142)	18	56.7 ± 1.7
Recreational athletes—men and women (157)	81	72.8 ± 9.1
National badminton—men (155)	12	73.3 ± 3.4

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

TABLE 14.15 Repetition Benchmarks for the 90-Second Box Jump Test

Age	14-15	16-17	18-20
FEMALE			
Target goal	73-75	80-85	82-89

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Good	67-72	75-79	76-82
Developing	51-66	60-74	70-75
Needs training	≤50	≤59	≤69
MALE			
Target goal	74-85	92-97	95-100
Good	67-73	85-91	87-94
Developing	62-66	73-84	75-86
Needs training	≤61	≤72	≤74

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TABLE 14.16 Percentiles by Age Groups and Sex for Partial Curl-Up

Percentile*	Age and sex									
	20-29		30-39		40-49		50-59		60-69	
	M	F	M	F	M	F	M	F	M	F
90	75	70	75	55	75	55	74	48	53	50
80	56	45	69	43	75	42	60	30	33	30
70	41	37	46	34	67	33	45	23	26	24
60	31	32	36	28	51	28	35	16	19	19
50	27	27	31	21	39	25	27	9	16	13
40	24	21	26	15	31	20	23	2	9	9
30	20	17	19	12	26	14	19	0	6	3
20	13	12	13	0	21	5	13	0	0	0
10	4	5	0	0	13	0	0	0	0	0

*Descriptors for percentile rankings: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average.

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TABLE 14.17 Fitness Categories by Age Groups and Sex for Push-Ups

Category	Age and sex									
	20-29		30-39		40-49		50-59		60-69	
	M	F	M	F	M	F	M	F	M	F
Excellent	36	30	30	27	25	24	21	21	18	17
Very good	35	29	29	26	24	23	20	20	17	16
	29	21	22	20	17	15	13	11	11	12
Good	28	20	21	19	16	14	12	10	10	11
	22	15	17	13	13	11	10	7	8	5
Fair	21	14	16	12	12	10	9	6	7	4
	17	10	12	8	10	5	7	2	5	2

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Needs improvement	16	9	11	7	9	4	6	1	4	1
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TABLE 14.18 Normative Data for the AFT Hand-Release Push-Up

	17-21		22-26		27-31		32-36		37-41		42-46		47-51		52-56		57-61		Over 61		
POINTS	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	POINTS
100	57	53	61	50	62	48	60	47	59	41	56	36	55	35	51	30	46	24	43	24	100
99	56	50	60	49	59	45	59	46	57	39	55	34	53	34	48	28	43	23	41	23	99
98	54	47	57	46	56	42	56	43	54	36	53	31	50	31	45	26	40	22	39	22	98
97	53	45	54	43	55	39	54	40	53	34	50	30	47	30	43	25	38	21	37	21	97
96	51	43	53	42	53	38	53	37	51	33	48	28	44	27	40	24	37	20	35	20	96
95	50	42	52	39	52	36	52	36	48	31	46	26	42	25	38	23	35	19	34	19	95
94	49	39	50	37	51	35	50	35	46	28	44	24	41	24	35	22	34	18	33	18	94
93	48	--	49	36	49	34	48	34	45	27	43	23	39	23	34	20	33	--	31	--	93
92	47	38	48	35	48	33	47	32	44	26	42	22	38	22	33	19	31	17	30	17	92
91	46	37	47	34	47	32	45	31	43	23	41	21	36	21	32	18	30	16	29	16	91
90	45	36	46	33	46	30	44	30	42	22	38	20	34	20	31	17	29	15	26	15	90
89	44	35	45	32	45	28	43	28	41	--	37	--	33	19	30	--	26	14	24	14	89
88	43	34	44	31	44	27	42	26	39	21	36	19	--	18	28	16	25	--	--	--	88
87	42	33	43	30	43	26	41	25	38	--	34	18	32	--	27	15	24	--	23	--	87
86	41	32	42	28	42	25	40	--	37	20	33	17	31	17	26	--	23	--	--	13	86
85	--	31	41	27	41	--	39	24	34	--	32	16	30	16	25	14	--	--	22	--	85
84	40	30	40	26	40	24	38	--	33	19	--	--	29	15	24	--	22	13	21	--	84
83	39	29	38	25	38	23	36	23	32	18	31	15	28	14	23	--	21	--	20	--	83
82	38	28	37	24	37	22	34	--	--	17	30	--	27	--	--	13	20	--	19	--	82
81	--	--	--	--	36	--	33	22	31	--	29	14	25	13	22	--	19	--	18	--	81
80	37	27	35	--	35	21	--	19	30	16	28	--	23	--	21	--	18	--	17	--	80
79	36	26	34	23	34	20	32	--	29	15	27	13	22	--	20	--	--	--	16	12	79
78	35	25	33	22	33	19	--	18	28	14	26	--	--	--	19	--	17	--	--	--	78
77	34	24	32	21	32	18	31	--	27	--	25	--	21	--	18	--	--	12	15	--	77
76	33	23	31	19	--	17	30	--	26	13	23	--	--	--	17	--	16	--	14	--	76
75	32	22	--	--	31	--	29	17	25	--	22	--	20	--	--	--	15	--	--	--	75
74	--	21	--	18	30	16	28	16	24	--	21	--	19	12	16	12	--	--	13	--	74
73	31	--	30	--	29	--	26	15	23	--	20	12	18	--	--	--	14	--	--	--	73
72	30	20	27	17	27	15	24	14	22	12	--	--	17	--	15	--	--	--	--	--	72
71	29	19	26	16	26	--	23	--	21	--	19	--	16	--	14	--	13	--	12	--	71
70	28	18	25	15	23	14	22	13	20	--	18	--	15	--	--	--	--	--	--	--	70

Reprinted from U.S. Army, "Army Fitness Test Score Tables: Hand-release Push-Up (HRP) (number of correctly performed repetitions in 2 minutes)," last modified June 1, 2025. **Before consulting the above table, check for the latest update** at: https://www.army.mil/e2/downloads/rv7/aft/AFT_Scoring_Scales_250601.pdf.

69	27	17	22	14	21	--	21	--	19	--	17	--	14	--	13	--	--	--	11	69
68	25	16	21	13	--	13	20	--	17	--	16	--	13	--	--	--	12	--	11	68
67	24	--	--	--	--	--	19	--	16	11	15	11	--	11	12	11	--	11	--	67
66	23	15	20	--	20	--	17	12	15	--	14	--	12	--	--	--	--	--	--	66
65	22	14	17	12	18	12	16	--	14	--	13	--	--	--	--	--	11	--	--	65
64	20	13	16	--	16	--	14	11	13	--	12	--	--	--	11	--	--	--	--	64
63	17	--	14	--	14	11	13	--	12	--	--	--	11	--	--	--	--	--	--	63
62	16	12	13	11	12	--	12	--	--	--	11	--	--	--	--	--	--	--	--	62
61	13	11	12	--	11	--	11	--	11	--	--	--	--	--	--	--	--	--	--	61
60	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10	60

Reprinted from U.S. Army, "Army Fitness Test Score Tables: Hand-release Push-Up (HRP) (number of correctly performed repetitions in 2 minutes)," last modified June 1, 2025. **Before consulting the above table, check for the latest update** at: https://www.army.mil/e2/downloads/rv7/aft/AFT_Scoring_Scales_250601.pdf.

TABLE 14.19 YMCA Bench Press Norms

Percentile*	Age and sex											
	18-25		26-35		36-45		46-55		56-65		>65	
	M	F	M	F	M	F	M	F	M	F	M	F
90	44	42	41	40	36	33	28	29	24	24	20	18
80	37	34	33	32	29	28	22	22	20	20	14	14
70	33	28	29	28	25	24	20	18	14	14	10	10
60	29	25	26	24	22	21	16	14	12	12	10	8
50	26	21	22	21	20	17	13	12	10	9	8	6
40	22	18	20	17	17	14	11	9	8	6	6	4
30	20	16	17	14	14	12	9	7	5	5	4	3
20	16	12	13	12	10	8	6	5	3	3	2	1
10	10	6	9	6	6	4	2	1	1	1	1	0

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

Score is the number of repetitions completed in 1 minute using an 80-pound (36 kg) barbell for men and a 35-pound (16 kg) barbell for women.

Adapted from YMCA (2000).

TABLE 14.20 Descriptive Data for the Star Excursion Balance Test

Score	High school	Collegiate
ANT (%)	69.9 ± 7.9	70.3 ± 7.1
PM (%)*	83.5 ± 10.2	86.7 ± 10.7

ANT = Anterior

PM = Posteromedial

PL = Posterolateral

COMP = Composite

*Statistically significant difference between groups ($N \leq 0.05$)

Reprinted by permission from R.S. McCann, K.B. Kosik, M.Q. Beard, et al., "Variations in Star Excursion Balance Test Performance Between High School and Collegiate Football Players," *The Journal of Strength and Conditioning Research* 29, no. 10 (2015): 2765-2770.

PL (%)*	72.8 ± 11.4	77.1 ± 10.2
COMP (%)*	75.4 ± 8.5	78.0 ± 7.4

ANT = Anterior

PM = Posteromedial

PL = Posterolateral

COMP = Composite

*Statistically significant difference between groups ($N \leq 0.05$)

Reprinted by permission from R.S. McCann, K.B. Kosik, M.Q. Beard, et al., "Variations in Star Excursion Balance Test Performance Between High School and Collegiate Football Players," *The Journal of Strength and Conditioning Research* 29, no. 10 (2015): 2765-2770.

TABLE 14.21 Normalized Reach Distance on the Star Excursion Balance Test for Each Limb, by Team

Sex/Sport	Anterior	Posterolateral	Posteromedial	Composite	Anterior	Posterolateral	Posteromedial	Composite
WOMEN	DOMINANT LIMB				NONDOMINANT LIMB			
Basketball	64.3 ± 6.9	88.7 ± 8.0	100.2 ± 7.1	84.4 ± 6.5	63.4 ± 7.9	85.7 ± 8.4	101.5 ± 7.7	83.5 ± 7.0
	(61.2, 67.4)	(85.1, 92.4)	(97.0, 103.5)	(81.5, 87.4)	(59.8, 67.0)	(81.8, 89.6)	(98.0, 105.0)	(80.3, 86.7)
Golf	69.4 ± 5.5	86.5 ± 10.2	105.3 ± 10.0	87.0 ± 7.7	67.6 ± 4.4	85.9 ± 8.7	102.2 ± 7.8	85.2 ± 6.2
	(65.4, 73.3)	(79.2, 93.8)	(98.2, 112.4)	(81.5, 92.6)	(64.5, 70.8)	(79.7, 92.1)	(96.7, 107.8)	(80.8, 89.7)
Hockey	66.6 ± 7.3	93.1 ± 10.7	110.3 ± 10.7	90.0 ± 8.6	67.0 ± 6.9	93.0 ± 11.0	113.1 ± 7.6	91.0 ± 7.5
	(64.1, 69.1)	(89.5, 96.8)	(106.7, 114.0)	(87.1, 92.9)	(64.6, 69.3)	(89.2, 96.7)	(110.5, 115.6)	(88.5, 93.5)
Soccer	66.8 ± 5.3	88.7 ± 8.8	104.3 ± 8.0	86.6 ± 6.3	67.4 ± 5.2	90.2 ± 9.9	105.3 ± 8.1	87.7 ± 6.7
	(65.1, 68.5)	(85.9, 91.5)	(101.8, 106.8)	(84.6, 88.6)	(65.8, 69.1)	(89.1, 93.3)	(102.8, 107.9)	(85.5, 89.8)
Softball	63.5 ± 5.6	87.0 ± 11.3	100.7 ± 8.8	83.7 ± 7.5	63.9 ± 4.8	85.3 ± 9.2	99.6 ± 9.3	82.9 ± 6.3
	(61.2, 65.9)	(82.4, 91.6)	(97.1, 104.3)	(80.6, 86.9)	(61.9, 65.8)	(81.5, 89.1)	(95.7, 103.4)	(80.3, 85.5)
Volleyball	66.8 ± 7.1	86.6 ± 10.4	102.7 ± 10.5	85.4 ± 8.5	65.3 ± 5.4	88.4 ± 9.5	102.3 ± 7.9	85.3 ± 6.9
	(63.6, 69.9)	(82.0, 91.2)	(98.1, 107.4)	(81.6, 89.2)	(62.9, 67.7)	(84.2, 92.6)	(98.8, 105.9)	(82.3, 88.4)
MEN	DOMINANT LIMB				NONDOMINANT LIMB			
Basketball	61.8 ± 4.3	86.6 ± 11.6	99.2 ± 7.8	82.5 ± 5.9	61.6 ± 4.8	86.9 ± 13.0	97.9 ± 7.9	82.1 ± 6.5
	(59.6, 64.1)	(80.6, 92.5)	(95.2, 103.2)	(79.5, 85.6)	(59.2, 64.1)	(80.2, 93.6)	(93.9, 102.0)	(78.8, 85.5)

*Values are mean ± SD (95% confidence interval) percent limb length.

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Golf	65.1 ± 6.2	83.9 ± 12.1	103.7 ± 9.0	84.2 ± 8.5	68.2 ± 4.7	85.3 ± 10.2	106.0 ± 7.2	86.5 ± 5.9
	(61.2, 69.1)	(76.2, 92.6)	(98.0, 109.4)	(78.9, 89.6)	(65.2, 71.2)	(78.9, 91.8)	(101.4, 110.6)	(82.8, 90.2)
Hockey	67.3 ± 5.2	95.6 ± 9.5	110.9 ± 5.5	91.3 ± 5.0	67.6 ± 4.0	96.6 ± 10.6	111.2 ± 5.7	91.8 ± 5.6
	(65.6, 69.0)	(92.5, 98.7)	(109.2, 112.7)	(89.6, 92.9)	(66.3, 68.9)	(93.2, 100.1)	(109.4, 113.0)	(90.0, 93.6)
Soccer	62.7 ± 4.9	89.9 ± 7.7	103.5 ± 8.6	85.4 ± 5.5	62.9 ± 5.0	89.9 ± 8.3	104.2 ± 8.8	85.7 ± 5.9
	(61.1, 64.2)	(87.5, 92.4)	(100.8, 106.2)	(83.6, 87.1)	(61.3, 64.5)	(87.3, 92.5)	(101.4, 107.0)	(83.8, 87.5)
Football	62.2 ± 8.2	88.6 ± 12.9	99.8 ± 8.7	83.5 ± 7.9	61.9 ± 7.3	87.9 ± 13.0	99.8 ± 8.9	83.2 ± 8.0
	(60.4, 64.0)	(85.8, 91.5)	(97.9, 101.7)	(81.8, 85.3)	(60.3, 63.5)	(85.1, 90.8)	(97.9, 101.8)	(81.4, 85.0)
Wrestling	66.7 ± 6.7	91.1 ± 11.9	110.3 ± 11.1	89.4 ± 8.2	67.8 ± 8.2	90.8 ± 10.3	111.0 ± 12.3	89.9 ± 8.3
	(64.7, 68.7)	(87.6, 94.6)	(107.0, 113.5)	(87.0, 91.8)	(65.4, 70.2)	(87.8, 93.8)	(107.4, 114.6)	(87.4, 92.3)

*Values are mean ± SD (95% confidence interval) percent limb length.

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TABLE 14.22 Maximal Rate of Oxygen Consumption, 1.5-Mile (2.4 km) Run Time, Sit-and-Reach Test, and Body Composition: Percentile Rankings for 20- to 29-Year-Old Males

Percentile ^a	$\dot{V}O_2\text{max}$ (mL · kg ⁻¹ · min ⁻¹)	1.5-mile (2.4 km) run time, min:s	Sit-and-reach test ^b		
			in.	cm	Body fat (%)
99	60.5	8:29			4.2
90	54.0	9:34	22	55.9	7.9
80	51.1	10:09	20	50.8	10.5
70	47.5	10:59	19	48.3	12.6
60	45.6	11:29	18	45.7	14.8
50	43.9	11:58	17	43.2	16.6
40	41.7	12:38	15	38.1	18.6
30	39.9	13:15	14	35.6	20.7
20	38.0	14:00	13	33.0	23.3
10	34.7	15:30	11	27.9	26.6
01	26.5	20:58			33.4

^aDescriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

^bSit and reach for 18- to 25-year-old men.

Adapted from American College of Sports Medicine (2014).

TABLE 14.23 Maximal Rate of Oxygen Consumption, 1.5-Mile (2.4 km) Run Time, Sit-and-Reach Test, and Body Composition: Percentile Rankings for 20- to 29-Year-Old Females

Percentile ^a	$\dot{V}O_2\text{max}$ (ml · kg ⁻¹ · min ⁻¹)	1.5-mile (2.4 km) run time, min:s	Sit-and-reach test ^b		Body fat (%)
			in.	cm	
99	54.5	9:30			11.4
90	46.8	11:10	24	61.0	15.1
80	43.9	11:58	22	55.9	16.8
70	41.1	12:51	21	53.3	18.4
60	39.5	13:24	20	50.8	19.8
50	37.8	14:04	19	48.3	21.5
40	36.1	14:50	18	45.7	23.4
30	34.1	15:46	17	43.2	25.5
20	32.3	16:46	16	40.6	28.2
10	29.5	18:33	14	35.6	33.5
01	23.7	23:58			38.6

^aDescriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

^bSit and reach for 18- to 25-year-old women.

Adapted from American College of Sports Medicine (2014).

TABLE 14.24 Maximal Rate of Oxygen Consumption, 1.5-Mile (2.4 km) Run Time, Sit-and-Reach Test, and Body Composition: Percentile Rankings for 30- to 39-Year-Old Males

Percentile ^a	$\dot{V}O_2\text{max}$ (ml · kg ⁻¹ · min ⁻¹)	1.5-mile (2.4 km) run time, min:s	Sit-and-reach test ^b		Body fat (%)
			in.	cm	
99	58.3	8:49			7.3
90	51.7	10:01	21	53.3	12.4
80	48.3	10:46	19	48.3	14.9
70	46.0	11:22	18	45.7	16.8
60	44.1	11:54	17	43.2	18.4
50	42.4	12:24	15	38.1	20.0
40	40.7	12:58	14	35.6	21.6
30	38.7	13:44	13	33.0	23.2
20	36.7	14:34	11	27.9	25.1
10	33.8	15:57	9	22.9	27.8
01	26.5	20:58			34.4

^aDescriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

^bSit and reach for 26- to 35-year-old men.

Adapted from American College of Sports Medicine (2014).

TABLE 14.25 Maximal Rate of Oxygen Consumption, 1.5-Mile (2.4 km) Run Time, Sit-and-Reach Test, and Body Composition: Percentile Rankings for 30- to 39-Year-Old Females

Percentile ^a	$\dot{V}O_2\text{max}$ (ml · kg ⁻¹ · min ⁻¹)	1.5-mile (2.4 km) run time, min:s	Sit-and-reach test ^b		Body fat (%)
			in.	cm	
99	52.0	9:58			11.2
90	45.3	11:33	23	58.4	15.5
80	42.4	12:24	22	55.9	17.5
70	39.6	13:24	21	53.3	19.2
60	37.7	14:08	20	50.8	21.0
50	36.7	14:34	19	48.3	22.8
40	34.2	15:43	17	43.2	24.8
30	32.4	16:42	16	40.6	26.9
20	30.9	17:38	15	38.1	29.6
10	28.0	19:43	13	33.0	33.6
01	22.9	24:56			39.0

^aDescriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

^bSit and reach for 26- to 35-year-old women.

Adapted from American College of Sports Medicine (2014).

TABLE 14.26 $\dot{V}O_2\text{max}$ Descriptive Data for Athletes in Various Sports

Category	Typical $\dot{V}O_2\text{max}$ of athletes playing the sport (ml · kg ⁻¹ · min ⁻¹)		Sport
	Males	Females	
Extremely high	70+	60+	Cross-country skiing Middle-distance running Long-distance running
Very high	63-69	54-59	Bicycling Rowing Racewalking
High	57-62	49-53	Soccer Middle-distance swimming Canoe racing Handball Racquetball Speedskating Figure skating Downhill skiing Wrestling
Above average	52-56	44-48	Basketball Ballet dancing American football (offensive, defensive backs) Gymnastics Hockey Horse racing (jockey)

Data from Nieman (1995).

			Sprint swimming Tennis Sprint running Jumping
Average	44-51	35-43	Baseball, softball American football (linemen, quarterbacks) Shot put Discus throw Weightlifting Bodybuilding

Data from Nieman (1995).

TABLE 14.27 Percentile Ranks for the 12-Minute Run

Percentile*	Age (years) and distance					
	20-29		30-39		40-49	
	km	miles	km	miles	km	miles
MEN						
90	2.90	1.80	2.82	1.75	2.72	1.69
80	2.78	1.73	2.67	1.66	2.57	1.60
70	2.62	1.63	2.56	1.59	2.46	1.53
60	2.54	1.58	2.48	1.54	2.40	1.49
50	2.46	1.53	2.40	1.49	2.30	1.43
40	2.37	1.47	2.32	1.44	2.22	1.38
30	2.29	1.42	2.24	1.39	2.14	1.33
20	2.20	1.37	2.14	1.33	2.06	1.28
10	2.06	1.28	2.01	1.25	1.95	1.21
WOMEN						
90	2.59	1.61	2.53	1.57	2.43	1.51
80	2.46	1.53	2.40	1.49	2.27	1.41
70	2.35	1.46	2.27	1.41	2.20	1.37
60	2.27	1.41	2.19	1.36	2.11	1.31
50	2.20	1.37	2.14	1.33	2.04	1.27
40	2.12	1.32	2.04	1.27	1.96	1.22
30	2.03	1.26	1.95	1.21	1.90	1.18
20	1.95	1.21	1.88	1.17	1.82	1.13
10	1.82	1.13	1.75	1.09	1.70	1.05

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

Adapted from American College of Sports Medicine (2014).

TABLE 14.28 Descriptive Data* for Yo-Yo Intermittent Recovery Test for Various Populations

Group, sport, or position	Number of athletes	Yo-Yo IR1
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*The values listed are means $\pm$ standard deviation. The data should be regarded as only descriptive, not normative.

		Distance (m)	Distance (yd)
National soccer—men (106)	18	2,260 ± 80	2,472 ± 87
National soccer—men (106)	24	2,040 ± 60	2,231 ± 66
National rugby league—men (4)	23	1,656 ± 403	1,811 ± 441
Semi-professional rugby league—men (4)	27	1,564 ± 415	1,710 ± 454
National soccer—women (108)	17	1,224 ± 255	1,339 ± 279
National soccer—men (108)	17	2,414 ± 456	2,640 ± 499
National junior soccer—women (108)	17	826 ± 160	903 ± 175
National junior soccer—men (108)	17	2,092 ± 260	2,288 ± 284
Under 17 soccer—men (37)	60	1,556 ± 478	1,702 ± 523
Under 16 rugby union—men (117)	150	1,150 ± 403	1,258 ± 441
Under 14 elite basketball—men (152)	15	1,100 ± 385	1,203 ± 421
Under 15 elite basketball—men (152)	15	1,283 ± 461	1,403 ± 504
Under 17 elite basketball—men (152)	17	1,412 ± 245	1,544 ± 268
Under 18 Gaelic football—men (34)	265	1,465 ± 370	1,602 ± 405

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

FIGURE 14.34 30-15 IFT performance among athletes of different ages, sports, and playing standards: (a) male and (b) female. Adapted by permission from Buchheit, M, Dikmen, U, and Vassallo, C. The 30-15 intermittent fitness test—two decades of learnings. *SPSR* 148:1-13, 2021.

TABLE 14.29 Descriptive Data* for Change-of-Direction Tests for Various Populations

Group, sport, or position	Number of athletes	Time (s)			
		5-10-5	T-test	5-0-5	Hexagon
College soccer—women (154)	51	4.88 ± 0.20			
High school soccer—women (154)	83	4.91 ± 0.22			
College lacrosse—women (154)	79	4.99 ± 0.24			
National softball—women (112)	10			2.66 ± 0.14	
National basketball—women (140)	12			2.69 ± 0.28	
College students—women (130)	34		11.92 ± 0.52		
College students—men (130)	52		10.08 ± 0.46		
NCAA Division II college soccer—men (93)	12	4.80 ± 0.33			
Recreational athletes—women (143)	20	5.23 ± 0.25	11.70 ± 0.67		
Recreational athletes—men (143)	24	4.67 ± 0.21	10.31 ± 0.46		
High school rugby league—men (142)	70			2.42 ± 0.12	
High school rugby league—men (50)	28			2.30 ± 0.13	
High school rugby league—men (147)	302			2.49 ± 0.14	
High school rugby league—men (147)	870			2.51 ± 0.15	
Junior national volleyball—men (47)	14		9.90 ± 0.17		
Junior national volleyball—women (47)	15		10.33 ± 0.13		
NCAA Division I college lacrosse—women (153)	84	4.99 ± 0.23			
NCAA Division III college lacrosse—women (73)	11	4.92 ± 0.22	10.50 ± 0.60		
MLB baseball—men (74)	62	4.42 ± 0.90			
AAA baseball—men (74)	52	4.53 ± 0.20			
AA baseball—men (74)	50	4.42 ± 0.68			
A baseball—men (74)	84	4.48 ± 0.54			
Rookie baseball—men (74)	90	4.54 ± 0.19			
High school volleyball—women (131)	27		10.96 ± 0.58		
NCAA Division I volleyball—women (131)	26		10.65 ± 0.52		
Recreational athletes—women (118)	52		12.52 ± 0.90		13.21 ± 1.68
Recreational athletes—men (118)	58		10.49 ± 0.89		12.33 ± 1.47
College athletes—women (118)	56		10.94 ± 0.60		12.87 ± 1.48
College athletes—women (118)	47		9.94 ± 0.50		12.29 ± 1.39

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

TABLE 14.30 Percentile Ranks for the 5-10-5 (Previously Pro Agility) Test (Seconds) in NCAA Division I College Athletes

Percentile*	Women's volleyball	Women's basketball	Women's softball	Men's basketball	Men's baseball	Men's American football
90	4.75	4.65	4.88	4.22	4.25	4.21
80	4.84	4.82	4.96	4.29	4.36	4.31
70	4.91	4.86	5.03	4.35	4.41	4.38
60	4.98	4.94	5.10	4.39	4.46	4.44
50	5.01	5.06	5.17	4.41	4.50	4.52
40	5.08	5.10	5.24	4.44	4.55	4.59
30	5.17	5.14	5.33	4.48	4.61	4.66
20	5.23	5.23	5.40	4.51	4.69	4.76
10	5.32	5.36	5.55	4.61	4.76	4.89
Mean	5.03	5.02	5.19	4.41	4.53	4.54
SD	0.20	0.26	0.26	0.18	0.23	0.27
<i>n</i>	81	128	118	97	165	869

*Descriptors for percentiles: 90 = well above average; 70 = above average; 50 = average; 30 = below average; 10 = well below average

Data collected using electronic timing devices. SD = standard deviation, *n* = number of athletes.

Adapted by permission from J. Hoffman, *Norms for Fitness, Performance, and Health* (Human Kinetics, 2006), 36-38, 55-58, 113.

TABLE 14.31 Descriptive Data for the Illinois Test: Boys 14 to 16 Years

Evaluation	Limits	%	Number of people included in the group
High	<15.83 s	2.28	0
Above average	15.83-16.93 s	13.59	4
Average	16.94-19.13 s	68.27	12
Below average	19.14-20.23 s	13.59	3
Low	>20.23 s	2.28	1

Reprinted by permission from Z. Boyan and V. Batchev, "Investigation on the 'Agility' Skill at 14- to 16-Year Old Students by Illinois Agility Test," *National Sports Academy 'Vasil Levski'* (2019): 444. <https://doi.org/10.37393/icass2019/82>.

TABLE 14.32 Descriptive Data for the Illinois Test: Boys and Girls 16 to 19 Years

Evaluation	Sex	
	Male	Female
Excellent	<15.2 s	<17.0 s
Above average	15.2-16.1 s	17.0-17.9 s
Average	16.2-18.1 s	18.0-21.7 s
Below average	18.2-18.3 s	21.8-23.0 s
Low	>18.3 s	>23.0 s

TABLE 14.33 Descriptive Data* for Speed Testing for Various Populations

Group, sport, or position	Number of athletes	Time (s)		
		10 m	20 m	40 m

*The values listed are means $\pm$ standard deviation. The data should be regarded as only descriptive, not normative.

College soccer—women (154)	51		3.38 ± 0.17	5.99 ± 0.29
High school soccer—women (154)	83		3.33 ± 0.15	5.94 ± 0.28
College lacrosse—women (154)	79		3.37 ± 0.14	5.97 ± 0.27
NCAA Division I college soccer—women (97)	15	2.31 ± 0.25		
Under 16 rugby league—men (149)	67	1.82 ± 0.07	3.13 ± 0.00	
Under 17 rugby league—men (149)	50	1.81 ± 0.06	3.12 ± 0.10	
Under 18 rugby league—men (149)	56	1.80 ± 0.06	3.09 ± 0.10	
Under 19 rugby league—men (149)	89	1.82 ± 0.07	3.11 ± 0.12	
Under 20 rugby league—men (149)	22	1.79 ± 0.06	3.07 ± 0.12	
National soccer—men (149)	17	1.82 ± 0.30	3.00 ± 0.30	
National rugby league—men (7)	20	1.61 ± 0.06		5.15 ± 0.02
State rugby league—men (7)	20	1.66 ± 0.06		5.13 ± 0.02
National rugby league forwards—men (30)	12	1.66 ± 0.20	3.00 ± 0.08	
National rugby league backs—men (30)	6	1.65 ± 0.15	2.91 ± 0.10	
State rugby league—men (50)	26	2.06 ± 0.18	3.36 ± 0.23	5.83 ± 0.31
National rugby union—men (33)	30	1.69 ± 0.10	2.93 ± 0.20	
National rugby league forwards—men (102)	63			5.27 ± 0.19
National rugby league backs—men (102)	55			5.08 ± 0.20
National badminton—men (155)	12	1.94 ± 0.18	3.35 ± 0.30	
National rugby league—men (48)	58	1.73 ± 0.07		5.25 ± 0.17
Under 18 elite rugby league—men (50)	28	1.81 ± 0.08	3.11 ± 0.12	5.56 ± 0.22
Under 18 sub-elite rugby league—men (50)	36	1.94 ± 0.11	3.28 ± 0.18	5.83 ± 0.35
High school rugby league—men (147)	302	1.88 ± 0.12	3.23 ± 0.16	
High school rugby league—men (147)	870	1.90 ± 0.12	3.27 ± 0.19	
Junior national volleyball—women (47)	20	1.90 ± 0.01		
Junior national volleyball—men (47)	14	1.80 ± 0.02		
Under 18 Australian rules football—men (164)	177		3.13 ± 0.09	
NCAA Division III college soccer—men (93)	12	1.96 ± 0.11		5.79 ± 0.31
NCAA Division I college soccer—men (136)	27	1.70 ± 0.10		4.90 ± 0.20
National Australian rules football—men (163)	35	1.89 ± 0.07	3.13 ± 0.10	5.40 ± 0.17
National Australian rules football—men (163)	30	1.70 ± 0.06	2.94 ± 0.08	
National soccer—women (65)	85	1.67 ± 0.07		
National soccer—women (65)	47	1.70 ± 0.07		
National soccer—women (1)	17		3.17 ± 0.03	
National soccer—men (141)	270	2.27 ± 0.40	3.38 ± 0.70	
Under 18 Gaelic football—men (34)	265		3.22 ± 0.15	
Group, sport, or position	Number of athletes	Time (s)		
		10 yd	20 yd	40 yd

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

NCAA Division I college lacrosse—women (153)		1.99 ± 0.10	3.37 ± 0.14	5.97 ± 0.26
MLB baseball—men (74)	62	1.52 ± 0.10		
AAA baseball—men (74)	52	1.55 ± 0.09		
NCAA Division I football—men (45)	281			4.88 ± 0.27
NCAA Division II football—men (45)	282			4.92 ± 0.26
NCAA Division III football—men (45)	205			4.96 ± 0.27

*The values listed are means ± standard deviation. The data should be regarded as only descriptive, not normative.

TABLE 14.34 Normative Data for the Balance Error Scoring System Test

Age	Women	Men
20-29	11.9 ± 5.1	10.4 ± 4.4
30-39	11.4 ± 5.6	11.5 ± 5.5
40-49	12.7 ± 6.9	12.4 ± 5.7
50-54	15.1 ± 8.2	13.6 ± 6.9
55-59	16.7 ± 8.2	16.4 ± 7.2
60-64	19.3 ± 8.8	17.2 ± 7.1
65-69	19.9 ± 6.6	20.0 ± 7.3

Adapted from Iverson and Koehle (2013).

TABLE 14.35 Equations for Calculating Estimated Body Density From Skinfold Measurements Among Various Populations

SKF sites ^a	Population subgroups	Sex	Age	Equation	Reference
Σ7SKF (chest + abdomen + triceps + subscapular + suprailiac + midaxilla + thigh)	Black or Hispanic	Women	18-55 years	Db (g/cc) ^b = 1.0970 – 0.00046971 (Σ7SKF) + 0.00000056 (Σ7SKF) ² – 0.00012828 (age)	Jackson et al. (82)
Σ7SKF (chest + abdomen + triceps + subscapular + suprailiac + midaxilla + thigh)	Black or athletes	Men	18-61 years	Db (g/cc) ^b = 1.1120 – 0.00043499 (Σ7SKF) + 0.00000055 (Σ7SKF) ² – 0.00028826 (age)	Jackson and Pollock (80)
Σ4SKF (triceps + anterior suprailiac + abdomen + thigh)	Athletes	Women	18-29 years	Db (g/cc) ^b = 1.096095 – 0.0006952 (Σ4SKF) – 0.0000011 (Σ4SKF) ² – 0.0000714 (age)	Jackson et al. (82)

^aΣSKF = sum of skinfolds (mm); Db = body density.

^bUse population-specific conversion formulas (see [table 14.36](#)) to calculate %BF from Db.

^cMale athletes = 1; female athletes = 0.

^dBlack athletes = 1; white athletes = 0.

Adapted by permission from A.L. Gibson, D.R. Wagner, and V.H. Heyward, *Advanced Fitness Assessment and Exercise Prescription*, 9th ed. (Human Kinetics, 2025), 273.

Σ3SKF (triceps + suprailiac + thigh)	White or anorexic	Women	18-55 years	Db (g/cc) ^b = 1.0994921 – 0.0009929 (Σ3SKF) + 0.0000023 (Σ3SKF) ² – 0.0001392 (age)	Jackson et al. (82)
Σ3SKF (chest + abdomen + thigh)	White	Men	18-61 years	Db (g/cc) ^b = 1.109380 – 0.0008267 (Σ3SKF) + 0.0000016 (Σ3SKF) ² – 0.0002574 (age)	Jackson and Pollock (82)
Σ2SKF (triceps + calf)	Black or white	Boys	6-17 years	%BF = 0.735 (Σ2SKF) + 1.0	Slaughter et al. (137)
Σ2SKF (triceps + calf)	Black or white	Girls	6-17 years	%BF = 0.610 (Σ2SKF) + 5.1	Slaughter et al. (137)
Suprailiac, triceps	Athletes	Women	High school and college age	Db (g/cc) ^b = 1.0764 – (0.00081 × suprailiac) – (0.00088 × triceps)	Sloan and Weir (138)
Thigh, subscapular	Athletes	Men	High school and college age	Db (g/cc) ^b = 1.1043 – (0.00133 × thigh) – (0.00131 × subscapular)	Sloan and Weir (138)
Σ3SKF (triceps + abdomen + thigh)	Athletes	Men or women	18-34 years	%BF = 8.997 + 0.24658 (Σ3SKF) – 6.343 (gender ^c) – 1.998 (race ^d)	Evans et al. (41)

^aΣSKF = sum of skinfolds (mm); Db = body density.

^bUse population-specific conversion formulas (see [table 14.36](#)) to calculate %BF from Db.

^cMale athletes = 1; female athletes = 0.

^dBlack athletes = 1; white athletes = 0.

Adapted by permission from A.L. Gibson, D.R. Wagner, and V.H. Heyward, *Advanced Fitness Assessment and Exercise Prescription*, 9th ed. (Human Kinetics, 2025), 273.

TABLE 14.36 Population-Specific Two-Component Model Formulas for Converting Body Density to Percent Body Fat

Population	Age (yr)	Sex	%BF ^a	FFB _d (g·cc ⁻¹) ^b
RACE OR ETHNICITY				
African American	9-17	Female	(5.24/Db) – 4.82	1.088
	19-45	Male	(4.86/Db) – 4.39	1.106
	24-79	Female	(4.85/Db) – 4.39	1.106
American Indian	18-62	Male	(4.97/Db) – 4.52	1.099

FFB_d = fat-free body density; Db = body density; %BF = percent body fat; NA = no data available for this population subgroup.

^aMultiply value by 100 to calculate %BF.

^bFFB_d based on average values reported in selected research articles.

Adapted by permission from A.L. Gibson, D.R. Wagner, and V.H. Heyward, *Advanced Fitness Assessment and Exercise Prescription*, 9th ed. (Human Kinetics, 2025), 273.

	18-60	Female	(4.81/Db) – 4.34	1.108
Japanese Native	18-48	Male	(4.97/Db) – 4.52	1.099
		Female	(4.76/Db) – 4.28	1.111
	61-78	Male	(4.87/Db) – 4.41	1.105
		Female	(4.95/Db) – 4.50	1.100
Singaporean (Chinese, Indian, Malay)		Male	(4.94/Db) – 4.48	1.102
		Female	(4.84/Db) – 4.37	1.107
White	8-12	Male	(5.27/Db) – 4.85	1.086
		Female	(5.27/Db) – 4.85	1.086
	13-17	Male	(5.12/Db) – 4.69	1.092
		Female	(5.19/Db) – 4.76	1.090
	18-59	Male	(4.95/Db) – 4.50	1.100
		Female	(4.96/Db) – 4.51	1.101
	60-90	Male	(4.97/Db) – 4.52	1.099
		Female	(5.02/Db) – 4.57	1.098
Hispanic	20-40	Male	NA	NA
		Female	(4.87/Db) – 4.41	1.105
ATHLETES				
Resistance trained	24 ± 4	Male	(5.21/Db) – 4.78	1.089
	35 ± 6	Female	(4.97/Db) – 4.52	1.099
Aerobic endurance trained	21 ± 2	Male	(5.03/Db) – 4.59	1.097
	21 ± 4	Female	(4.95/Db) – 4.50	1.100
All sports	18-22	Male	(5.12/Db) – 4.68	1.093
	18-22	Female	(4.97/Db) – 4.52	1.099
CLINICAL POPULATIONS				
Anorexia nervosa	15-44	Female	(4.96/Db) – 4.51	1.101
Obesity	17-62	Female	(4.95/Db) – 4.50	1.100
Spinal cord injury (paraplegic or quadriplegic)	18-73	Male	(4.67/Db) – 4.18	1.116
		Female	(4.70/Db) – 4.22	1.114

FFB_d = fat-free body density; Db = body density; %BF = percent body fat; NA = no data available for this population subgroup.

*Multiply value by 100 to calculate %BF.

^bFFB_d based on average values reported in selected research articles.

Adapted by permission from A.L. Gibson, D.R. Wagner, and V.H. Heyward, *Advanced Fitness Assessment and Exercise Prescription*, 9th ed. (Human Kinetics, 2025), 273.

TABLE 14.37 Percent Body Fat Descriptive Data for Athletes in Various Sports

Typical percent body fat of athletes playing the sport			
Classification	Males	Females	Sport

Data from D.C. Nieman, *Fitness and Sports Medicine: A Health-Related Approach*, 3rd ed. (Bull Pub. Co., 1995).